

Drilling and Blasting in Surface Mines

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Drill and blast operations have a unique place in mining and civil engineering. When the volume of hard rock mass to be excavated is very large, the use of mechanical cutting methods is snail-slow and ineffective. Blasting, however, can quickly fragment even the hardest known rocks over a large area. Breaking of rock mass into fragments requires energy. For any specific weight of rock mass the surface area is inversely proportional to the size of the fragments formed in fragmentation (Figure 1). Blasting forms the most economic method of rock fragmentation.

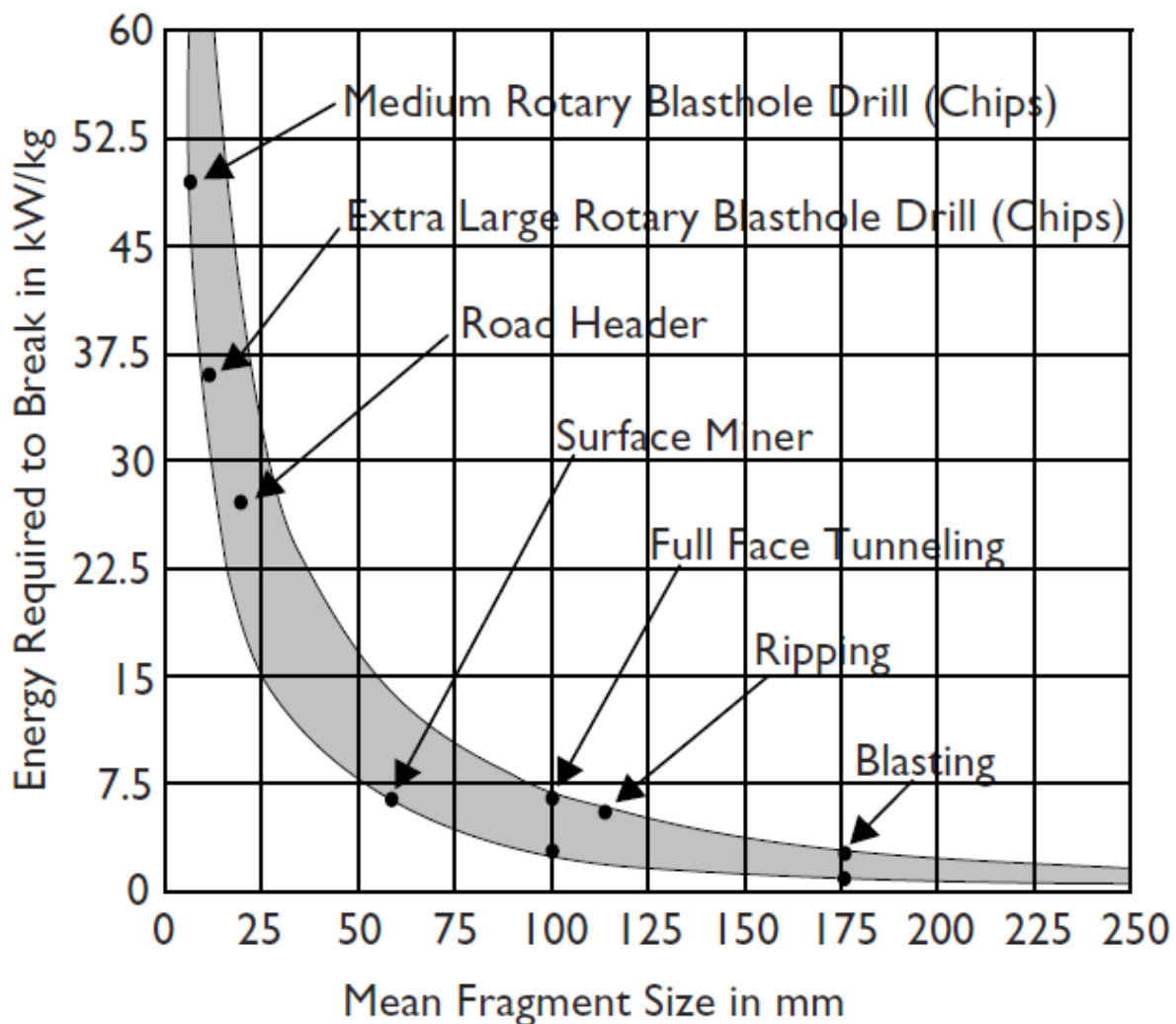


Figure 1: Relationship between energy requirement and mean fragment size for different excavation technologies (adapted from Gokhale 2011)

For a typical surface rock excavation, many blastholes are drilled systematically in a vertical or near vertical direction, filled with explosives and then blasted. These blastholes have specific diameter, depth and distance between them. After removal of the fragmented rock, depending upon the extent of excavation,

many step-like benches are formed. The nomenclature and terminology used for the purpose of drilling and blasting is shown in Figure 2.

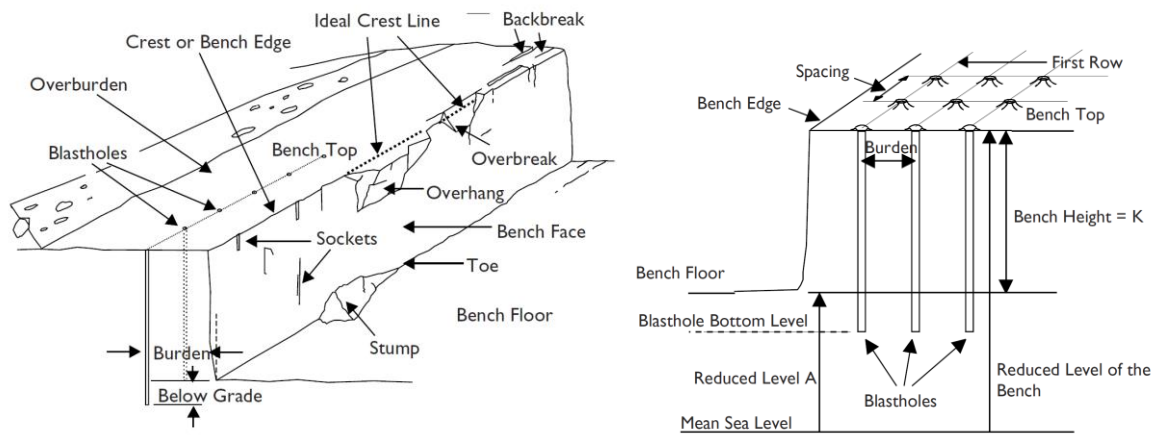


Figure 2: Nomenclature and terminology used for the purpose of drilling and blasting

When rock mass is blasted by the drill and blast method, the confinement of the explosive in the blastholes generates extremely high pressure and a huge quantity of heat. When this pressure and heat passes on to the surrounding rock mass, the rock mass just behind the wall of the blasthole gets pulverized into very fine particles. This happens because in that region new cracks are developed due to high energy concentration. However, at greater distances only the most vulnerable pre-existing cracks expand and join together to form a parting plane. Many such planes give rise to very large fragments in comparison to those formed by the other methods of breaking rocks.

Fundamentals and Technical Requirements for Drilling

In actual practice, drilling is carried out by using a drill bit connected at the bottom of a drill string that consists of drill pipes (Figure 3). A drill string often consists of some other items, not shown in the figure, that have specific purposes. The drilling bit is meant to interact with the rock. When energy in some form is passed to the rock from the drill bit the rock fractures. The drill string enables a continuous supply of energy to the drill bit so that rock fracture continues at progressive depths.

As the bit interacts with rock and breaks it, rock cuttings are formed. These cuttings are moved away by means of a circulating fluid that is pumped to the bit through the central bore in all the drill string components. When fluid escapes from the bit and starts moving up to the ground surface through the annular space, as shown in the figure, it drags the cuttings along with it and deposits them on the ground surface. Thus, the bit keeps on breaking fresh portions of the rock. The drill string is also given rotary motion so the bit rotates and its cutting points cover a complete cross sectional area to form a cylindrical blasthole.

The following technical requirements must be fulfilled for drilling process to commence, continue, and be complete:

1. The drilling bit, meant for breaking the rock formation, must have adequate hardness, heat resistance, strength, toughness, etc., so it can keep on breaking the formation over a long period of time.
2. Energy, in one of the many possible forms, must be supplied to the bit to cause rock disintegration.
3. The drill bit must be rotated to ensure a circular hole, irrespective of bit cross section.

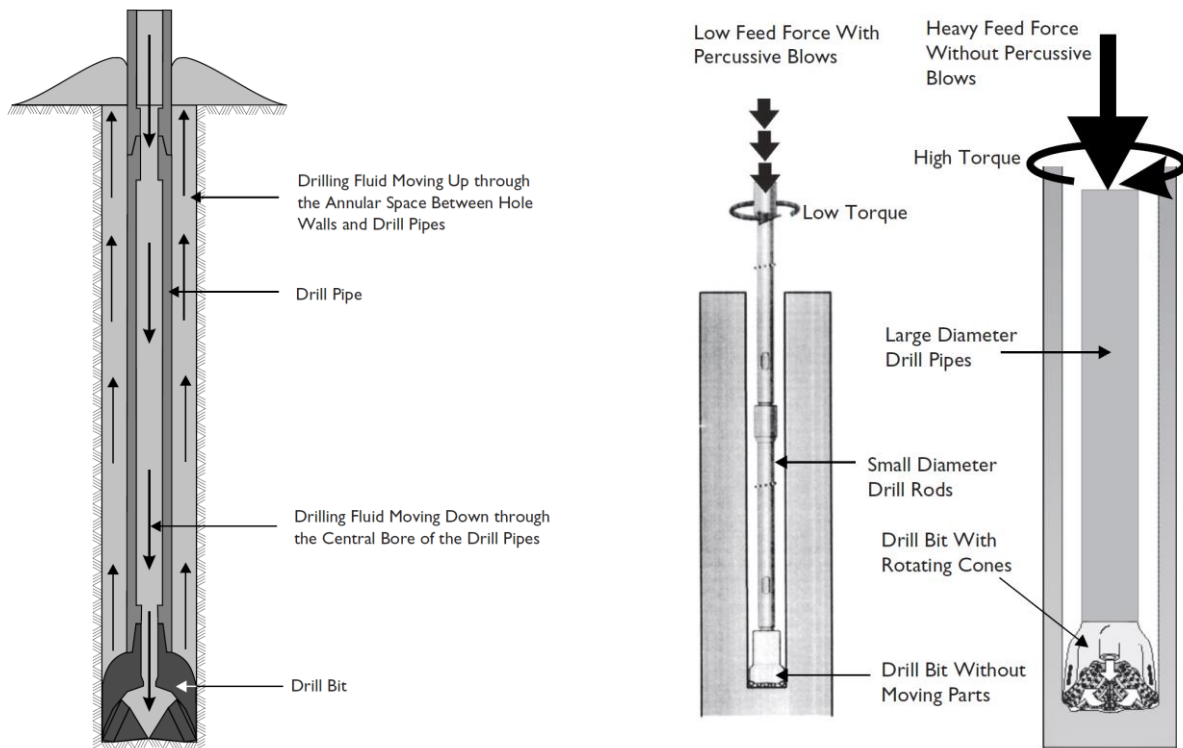


Figure 3: Schematics of drilling, percussive drilling and rotary drilling

4. Components in the drill string must also have specific properties to effectively transfer feed force, rotary torque, percussion blows etc., without undue deformations.
5. The rock cuttings formed in the process of formation disintegration must be taken out of the hole as soon as possible, so they do not waste energy in getting crushed once again.
6. The drill should be able to supply sufficient energy to the drill bit through the drill string components to cause formation fracture.
7. The drilling bit must be advanced in the intended direction of the drillhole.
8. The walls of the hole, formed as the drilling progresses, must be prevented from collapsing inside the hole.
9. After attaining the desired drillhole depth, it must be possible to withdraw the drill string composed of drill bit, drill pipes and other accessories.

Besides the above technical requirements, it is also necessary to fulfill certain other environmental and economic needs.

- a. Cuttings of very small size – in the form of dust – generated during the process of drilling, should be prevented from mixing with atmospheric air and polluting it.

- b. The process of drilling should continue at optimum speed so as to keep the cost of the operation to a bare minimum.
- c. Noise generated by the drilling equipment must be kept to an acceptable low level.

Two forms of mechanical energy, namely percussive and rotary, are used for drilling.

In either case the drill string and the drill bit are rotated. Accordingly, the drilling methods are called rotary percussion drilling and rotary drilling. A schematic of both these drilling methods is in Figure 3 above.

In rotary percussive drilling the energy generated at the drill is in the form of repeated blows like those of a hammer used in manual chiseling. These blows create shock waves in the drill string. The shock waves that travel to the bit are passed on to the rock mass through the cutting edges or points on the bit. As a result, cracks are formed in the rock mass and result in formation of rock chips.

In rotary drilling the drill pushes the bit into the rock mass with very heavy force. When this heavy feed force is transmitted to the rock mass through the cutting points of the drill bit, cracks develop and rock chips are formed.

In both rotary and rotary percussion drilling, cuttings are removed from the hole by a fluid circulated from bottom to top.

A drill contains many assemblies meant for various purposes. In a rotary percussion drill the assembly used for generating blows is called the hammer. Hammers are of two types viz. top hammers and down-the-hole hammers. A top hammer is so called because it always remains at the top of the drill string and hence out of the hole. It creates impacts by causing to and fro movement of a piston contained within it. In addition, it also generates rotary motion through a motor within it. The piston movement or rotary motion are caused by hydraulic fluid or compressed air (now obsolete) depending upon the type of drill.

Top hammers give fastest penetration rates in hard rocks. They are mostly used for drilling holes for different purposes on construction sites, small opencast mines, quarries and underground mines. The diameter of holes that can be drilled with top hammers is usually within 38 to 152 mm. The efficiency of top hammer drilling reduces rapidly as more drill rods are added to reach greater depth. Drilling holes deeper than 30 m with a top hammer is difficult because of the energy lost at the connections of the drill rods. Some top hammers are rated for drilling holes to 50 m depth by use of special drill string components.

A down-the-hole hammer, usually abbreviated as DTH hammer, is attached to the bottom of the drill string. A drill bit is attached to the lower end of the DTH hammer. A DTH hammer always goes inside the hole along with the bit. The movement of the piston in the DTH hammer is caused by compressed air. The rotary movement for drilling is generated by the rotary head in the drill and is passed on to the drill bit through drill string components and DTH hammer.

DTH hammers give fast penetration rates in hard rocks. They are also used for drilling holes for different purposes on construction sites, opencast mines, quarries and underground mines. The diameter of holes that are usually made with DTH hammers is within 100 to 200 mm. Many large hammers, used with specially built drills, are capable of drilling holes of diameter up to 900 mm and in case of specially built hammers even to 1500 mm. Holes of depth 1000 m or even more can be drilled by DTH drilling as the impact energy used for drilling is generated right above the bit and is, therefore, not lost in transmission.

Drill bits used with top or DTH hammers are made from alloy steel. On one side they have tungsten carbide inserts. For most of the bits used now, the inserts are in the form of buttons because such bits are found to be superior to chisel shaped inserts arranged in cross or X shape. On their other side, top hammer bits have

female threads that enable them to be attached to the drill rods. DTH bits have splines and collars for attachment to the hammers. Tungsten carbide insert buttons are shrink fitted in the holes made in the steel body, whereas the chisel-shaped inserts are brazed into pre-machined grooves by means of a suitable brazing alloy.

Top hammer drilling bits are small in size and are firmly coupled to the drill rods by means of threads. DTH bits are loosely coupled to the DTH hammers and have splines to ensure rotation as well as a collar at their top end to ensure that the bits are not detached from the hammer while drilling is carried out or when they are retracted after completion of the hole.

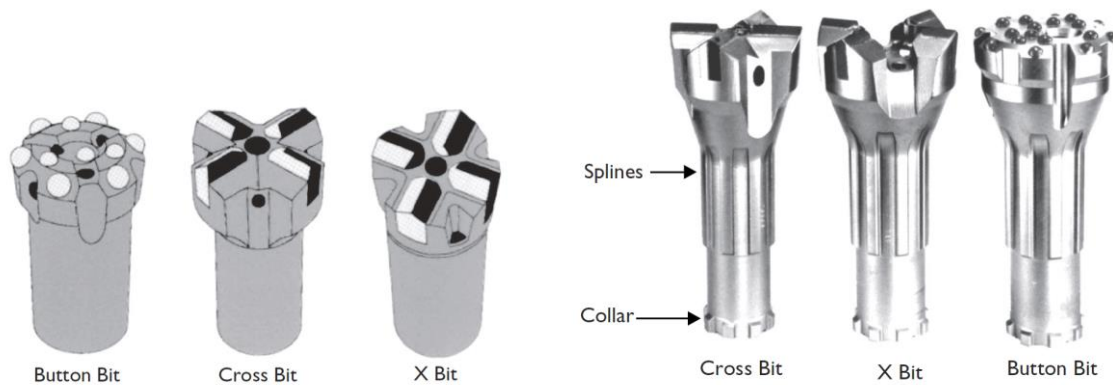


Figure 4: Bits used for rotary percussive drill machines

In a rotary blasthole drill, the rotary motion is generated at the drill head by hydraulic or electric motors. The head is pushed down through the feed mechanism to exert the necessary feed force. Drills used for rotary blasthole drilling are very large and heavy, as they have to exert a very high feed force and high torque on the bit through the drill string. Crawler mounting is best suited for the large, tall and heavy drills and for traveling in rough ground.

The components, from top to bottom, in a rotary drill string are a shock absorber, a saver sub, drill pipes, a stabilizer and a drill bit. Rotary drilling is carried out by using tricone roller bits in soft to very hard rock mass. In a soft rock mass, drag bits or claw bits are also used. These three types of bits are shown in Figure 5. All the three types of bits use tungsten carbide inserts fixed onto a steel body either by shrink fitting, brazing, or mechanical pin or circlip fitting. Tricone bits have three rotating cones fixed into the bit legs through roller and ball bearings. Unlike rotary percussion bits, they have moving parts.

Under the heavy force exerted onto the bit, the buttons or teeth on the rotating cones of the bit penetrate into the rock mass. With the heavy torque, they crush, scrape or plow the rock mass to create cuttings. These cuttings are moved out by drilling fluid circulated through the central bore of the drill string components.



Figure 5: Bits used for rotary drill machines

Table 1: Comparison of blasthole drilling methods on the basis of essential requirements

Criteria of comparison	Rotary percussion blasthole drilling		Rotary blasthole drilling
	Top hammer drilling	Down the hole hammer drilling	
Means of achieving formation disintegration	By means of blows generated at the drilling machine located above the ground.	By means of blows generated in the down the hole hammer placed just above the drill bit in the drill hole.	By means of static pressure given by the feed force exerted by the drilling machine placed on ground.
Type of drill string	Relatively slender as it comprises of drilling rods coupled together by means of sleeves.	Relatively rigid because it comprises of large diameter drill pipes that have male and female threads on their ends. No sleeve is required for coupling the pipes.	Rigid because it comprises of even larger diameter and heavier drill pipes that have male and female threads on their ends. No sleeve is required for coupling the pipes.
Transfer of impacts and rotary motion	Both impacts and rotary movement are transferred through drill string.	Only the rotary movement is transferred through drill string. Impacts are generated at the hammer in the hole just above the bit.	Very heavy static feed force instead of impacts and rotary movement are transferred through the drill string.

Criteria of comparison	Rotary percussion blasthole drilling		Rotary blasthole drilling
	Top hammer drilling	Down the hole hammer drilling	
Practical Hole Dia. Range	38 mm to 150 mm.	89 mm to 250 mm.	150 mm to 445 mm.
Practical hole depth range and susceptibility to hole deviation	0 to 18 m. Highly susceptible to deviation at higher depths due to high slenderness ratio of the drill rods.	0 to 100 m. Hole deviation is relatively small because drill pipes have low slenderness ratio.	0 to 60 m. Hole deviation is relatively small because drill pipes have low slenderness ratio.
Possibility of angle hole drilling	As holes are drilled with light equipment they can be drilled at any angle.	As holes are drilled with medium equipment drilling holes in direction other than vertical or near vertical is difficult.	Drilling equipment is very heavy. Vertical down or slightly inclined holes are possible.
Effect of depth on penetration rate	Penetration rate is considerably reduced with depth because of loss of percussion energy. It is absorption by drill rods.	No significant loss of penetration rate as the hole depth increases because no energy is lost in transmission.	Penetration rate is unaffected by increase in the depth of the hole because impact energy is not used.
Means of cutting removal from the hole	By means of compressed air circulated through the drill string. Sometimes water is also used as circulation medium.	By means of compressed air circulated through the drill string.	By means of compressed air circulated through the drill string.
Magnitude of feed force required on drilling tool	Small force is essential to ensure that the bit is in contact with the formation.	Small force is essential to ensure that the bit is in contact with the formation.	Very heavy force is essential to ensure that the bit remains in contact with the formation and crushes the formation beneath it.

Criteria of comparison	Rotary percussion blasthole drilling		Rotary blasthole drilling
	Top hammer drilling	Down the hole hammer drilling	
Effect of Extreme Cold Weather	As hydraulic oil is the medium of power transfer, great difficulties are experienced due to oil viscosity changes.	As compressed air is the medium of power transfer, difficulties in cold weather are easily surmountable.	Electric drills operate satisfactorily. In hydraulic drills difficulties are experienced due to oil viscosity changes.

CLASSIFICATION OF DRILLING

Energy in many different forms can be applied to a rock mass for creating a blasthole. To ensure that subsequent blasting is very effective, a blasthole must have some particular dimensional properties. As a blasthole is required to have set dimensions, energy has to be applied in a very controlled form. Three forms of energy can be applied to a rock mass in controlled manner. They are

- Mechanical Energy
- Heat Energy
- Chemical Energy

Based on these three forms of energy application, many drilling methods as shown in Figure 5a, are recognized. Of these three forms of energy, mechanical energy is almost exclusively used in actual drilling practice. The other two forms viz. heat and chemical energy are yet to prove their potential from an economic angle. Therefore, the methods that use mechanical energy for formation disintegration are called conventional drilling methods. Methods that use other two forms are non-conventional drilling methods.

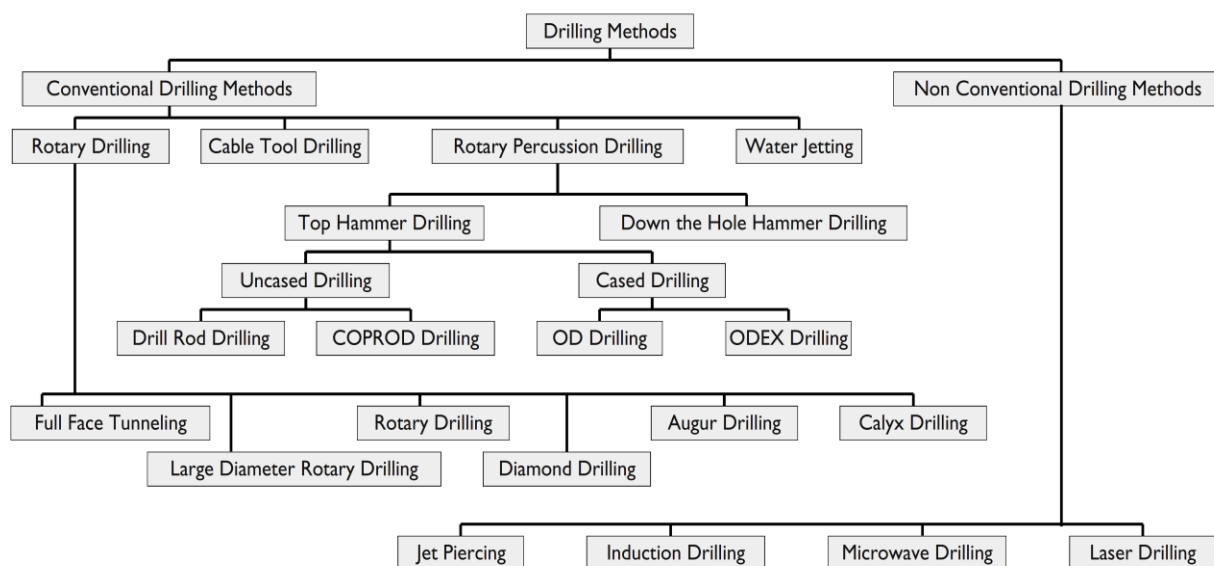


Figure 5a: Drilling methods

Choice of Drilling Machines

The main factors that influence the choice of drilling method are:

1. Period over which operations continues.
2. Diameter of the hole.
3. Depth of the hole.
4. Characteristics of the formation to be drilled.

In the case of civil engineering, the quantum of blasthole drilling to be carried out is relatively very small as compared to mining. A civil engineering contractor may rarely have a need for continuous blasthole drilling at one location for more than two years. Thus, for a civil contractor it is wise to select many smaller machines rather than few large machines for his civil engineering projects, because after completion of the project the smaller machines can be distributed on many separate small jobs.

Blastholes in civil engineering are rarely larger than 127 mm in diameter and almost never larger than 200 mm. The depth of a blastholes is also limited to about 20 m. Civil engineering firms, therefore, usually choose top hammer drilling for blastholes of smaller diameter and DTH drilling in case of larger blastholes.

In mining choosing the right drilling method is very critical as the blasthole drilling operations continue for decades and blasthole drills specifically designed for a method are to be procured before commencing the operation. Choice of incorrect drills can be disastrous as reselling drills at a fair price is difficult. When blasthole drilling is to be carried out in the soft sedimentary overburden of opencast coal mines, the diameters of the hole range between 152 to 381 mm and depths range between 15 to 55 m. In soft thick coal beds blasthole diameters range between 152 to 200 mm and depths between 10 to 20 m. For soft sedimentary formations suitability of a blasthole drilling method is illustrated in Figure 6.

In many opencast metal mines the formations are hard and the variation in hardness is over a very wide range. Here both overburden and ore are removed by using shovels and dumpers. For this reason the blasthole depths lie between 10 to 20 m and diameters can be from 100 to even 445 mm. For all such cases the blasthole drilling method can be chosen on the basis of the illustration in Figure 7.

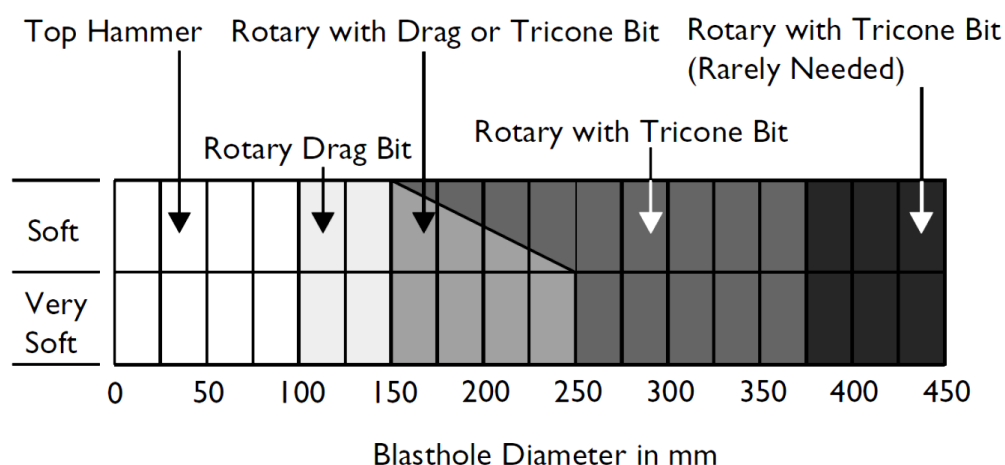


Figure 6: Suitability of blasthole drilling method in very soft and soft formations.

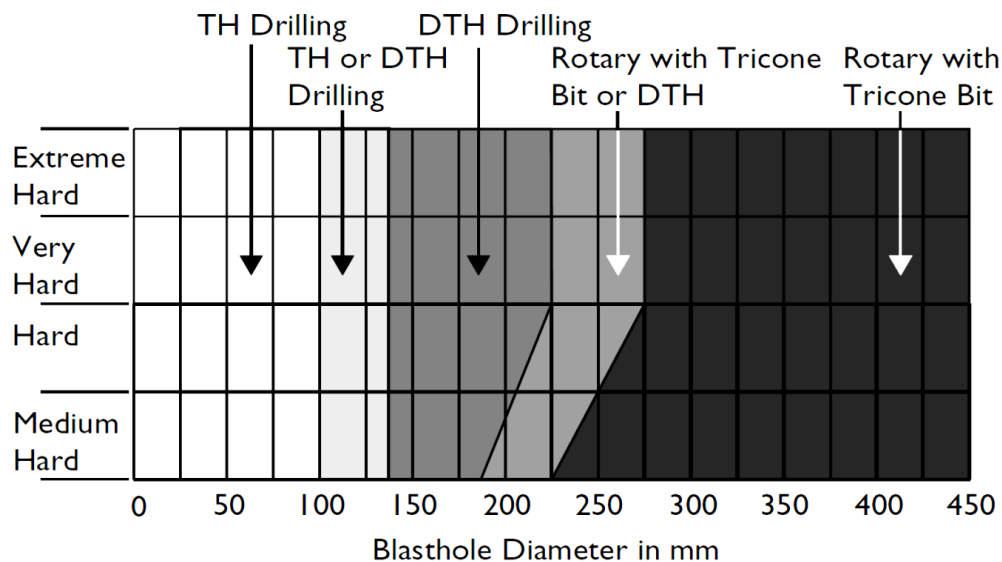


Figure 7: Suitability of blasthole drilling method in medium to very hard formations

In underground coal or metal mines, blasthole diameters are between 51 to 100 mm but in certain cases can be as high as 152 mm. The depths of blastholes can be from 2 to 30 m.

In coal mines it is very common to use rotary drag bit drilling. In hard coal formations top hammers are often used for drilling blastholes. When holes of diameter 152 mm to 30 m depth are to be drilled, as in some metal mines, DTH hammer is the most usual choice because the size of the drilling machines used for DTH drilling is small.

In tunneling or shaft sinking operations, diameters of blastholes are 38 to 64 mm. The depths of the blastholes, in a very few instances, can be as much as 25 m but in most of the cases depths are between 2 to 6 m. The top hammer rotary percussion drill is almost exclusively used in all such operations.

Two important points must always be remembered while choosing the right drill for blasthole drilling.

Rotary drilling requires exerting a heavy feed force in the downward direction on the drill string. Hence the drills are very heavy. If they are not heavy, their ability in drilling large diameter blastholes remains limited and must be investigated cautiously.

Percussive drills need not exert heavy feed force, but when drilling upward holes the weight of the piston acts adversely to the blow energy. Percussive drill performance must be judged in the light of this fact.

The performance of blasthole drilling equipment can be judged by means of carrying out certain laboratory tests. However, due to many influencing factors, it cannot be totally relied upon, while tackling an important issue like choosing a blasthole drill. Field trials by using hired equipment before taking a final decision about the suitable blasthole drilling method is a far better alternative, even if it is expensive.

Rotary Blasthole Drills

Rotary blasthole drills are divided into four classes, on the basis of the Feed Force (FF) they can exert on the drilling bit, as follows:

Small	100 kN < FF < 225 kN
Medium	226 kN < FF < 350 kN
Large	351 kN < FF < 500 kN
Extra Large	FF > 501 kN

List of manufactures of rotary blasthole drills showing the manufacturing range is given in Table 2.

Table 2: Rotary blasthole drill manufacturers and their range.

Manufacturer	Country of manufacture	Manufacturing range			
		Small	Medium	Large	E. large
Atlas Copco	USA	Yes	Yes	Yes	Yes
Bucyrus Intl.	USA	Yes	Yes	Yes	Yes
Changsha Keda CMCL	China	Yes	Yes	Yes	Yes
Emico Elecon	India	Yes	Yes		
Globe Drill	Australia			Yes	
Hausherr	Germany	Yes	Yes		
Heavy Eng. Corp.	India		Yes		
KLR	India	Yes	Yes		
LMP Precision	India	Yes	Yes		
NKMZ	USSR		Yes	Yes	
P&H Mining	USA			Yes	Yes
Revathi Equipment	India	Yes	Yes	Yes	
Rudgormash	USSR		Yes	Yes	
Sandvik Driltech	USA	Yes	Yes	Yes	Yes

All through the history of rotary blasthole drilling, the objective in improving machinery has been to introduce safety, efficiency, reduced maintenance and automation. The earliest step towards this was pipe handling. Carousel pipe changers were used even in the early 1950s. Single pipe changers, developed in the 1960s, were better at handling long and heavy drill pipes.

Drilling recorders were introduced in blasthole drills in the early 1960s. In the early 1980s they became electronic. Automatic fire extinguishing systems and automatic lubricating systems were incorporated into blasthole drills in the late 1960s.

In 1971 blasthole drills entered an era of automation, when Bucyrus Erie adapted a primary logic circuit to control some of the drilling operations on their blasthole drill model 60R.

Early electric drills featured Ward Leonard controls for accurately controlling the DC electric motors in the rotary head. Some manufacturers adopted AC motors and frequency variation to control the motors in the early 1980s. However, advances in Static or Thyristor Control for DC motors have still retained DC motors on the rotary heads of blasthole drills.

In the 1990s integrated circuits revolutionized the blasthole drills to a very great extent. They made it possible to precisely measure and monitor several functions in the drilling operation. Some features offered by such systems were useful in the following manner.

1. With the drilling recorder the driller could record the values of as many as six parameters of the drilling activities. With this facility the operators could monitor the drilling process very effectively. Later, when digital systems were introduced the number of parameters went to as much as 12 and their values could be seen on the screen in the operator's cab.
2. Blasthole drills could be automatically leveled within an accuracy of 0.25°.
3. With specialized drill process software, stored in the computerized system of the blasthole drill, it became possible to control drilling operations in the most appropriate manner by automatically adjusting feed force, feed rate, rotary speed, compressed air supply and so on.
4. Radio control for propelling operations of the blasthole drills made it possible to move the drill to the next blasthole by an operator who stood on the ground near the new hole position rather than the operators cab of the drill at the position of the already drilled hole. Very high positioning accuracy was achieved through such remote control propelling.
5. With the introduction of Global Positioning Systems (GPS) it became possible not only to navigate the blasthole drill even more accurately, but also to integrate the drilling operation with a comprehensive mine operation system to increase productivity.

RECENT ADVANCES IN DRILLING

Blasthole drilling is carried out in repetition of four main activities. These are,

1. Trimming
2. Leveling
3. Drilling
4. Pipe changing
- 5.

These four activities are the part of a cycle that repeats for drilling each of the thousands of blastholes. Full-fledged computerization of a rotary blasthole drill usually consists of four subsystems as follows:

1. Drilling Knowledge System
2. Drill Automation System
3. Drill Navigation System
4. Integrated Mining System

Each of these subsystems has a specific objective and is incorporated in a blasthole drill by using specific components. With exception of the integrated mining system, the other three subsystems are almost exclusive of each other in the sense that only one of the three can be present on a rotary blasthole drill. The objective of drilling knowledge systems, in short, is to get data about the drilling operation and the material being drilled. The objective of a drill automation system is to introduce automation in various activities to be carried out in a drilling cycle. Means of introducing automation in rod handling, leveling and drilling have been in existence for a long time. A drill navigation system aims at automating the trimming activity of the drilling cycle. Finally, the integrated mining system has an ultimate objective of controlling all the operations of all the equipment used in the mine. To achieve this, it is essential to channel some output from the drilling activities to the activities of other mining equipment.

To work properly and achieve the objective, there has to be communication between various components used for computerization. If only the blasthole drills are to be computerized to optimize drilling and blasting, the communication between the components is like the one shown through schematics in Figure 8.

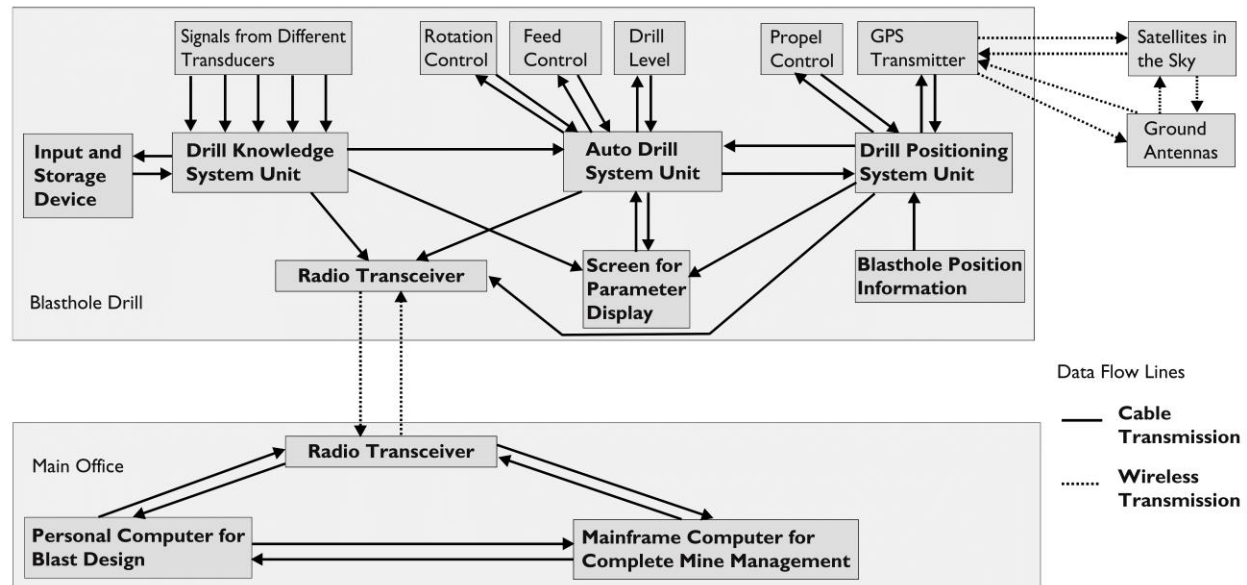


Figure 8: Schematics of data flow in rotary blasthole drill equipped with drill knowledge, auto drill and GPS positioning system

DRILLING KNOWLEDGE SYSTEMS

Right from the inception of drilling, the importance of knowledge about the formations encountered while drilling was recognized. Drillers always carefully watched the cuttings ejected with the flushing fluid and correlated the observations with the properties of the formation according to their experience. At the same time they also monitored the condition of the drill and drilling operations in general and controlled the feed force, rotary speed etc., on the basis of conditions visualized by them. Very few instruments such as rotary speed gage, ampere meters, air pressure gage, feed force indication gage etc. were available to them. Gradually recording and logging units were developed as explained below.

Formation logging units: Logging units that work on different principles have been developed. Some of these techniques are electric resistivity, neutral gamma, neutron, sonic, temperature, optical, caliper etc. These techniques are used for gathering rock mass data at great depths in the oilwells. Occasionally they have also been used for blasthole logging.

Analog Drilling Recorders: In the mid-1960s the recorders developed for oilwell operations were modified for top drive rotary blasthole drills. These blasthole drilling recorders recorded information only about drill operations. The recorders were pen type. They received data from transducers. The arms of the recorder moved as per the magnitudes of the data sensed by a transducer and plotted it on continuous paper that was moved at a constant speed. The data were usually converted into meaningful units shown on the continuous paper before being plotted. For example, hydraulic pressure in the feed force circuit was sensed but the plot showed feed force in terms of thousand pounds.

Electronic drilling recorders: One of the first digital drilling recorders was installed on a rotary blasthole drill in 1982. It stored data in the memory of the device. Since then digital drill recording devices have improved tremendously. Many manufacturers now make digital drill recording devices. They are also known as 'Drill Monitors' or 'Drill Efficiency Indicators'. A specimen output is shown in Figure 9.

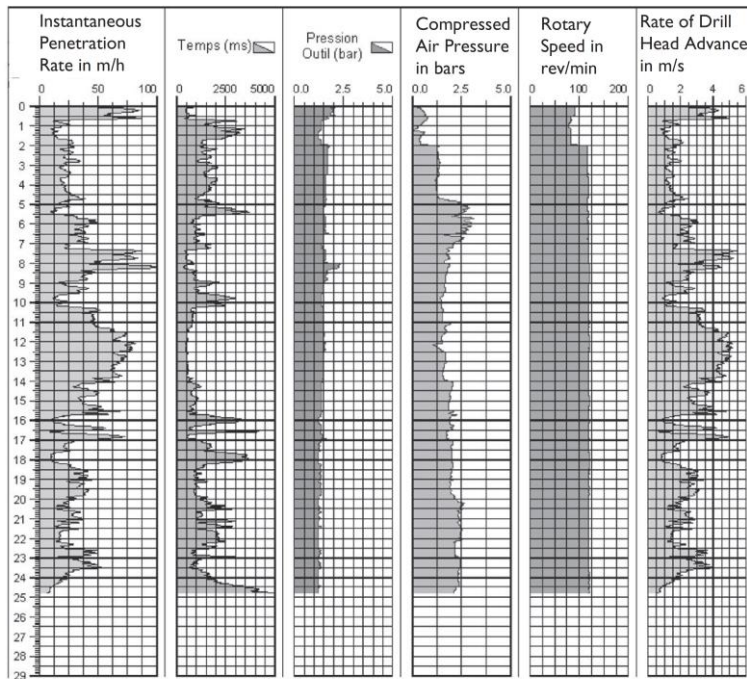


Figure 9: Drilling parameter record by digital recorder

POSITIONING BY GPS

GPS stands for Global Positioning System. GPS works by combining three segments viz.

- satellites in space,
- control stations on the ground that ensure precise positioning of the satellites with respect to the center of the earth, and
- the information fed by the user.

The satellites available for positioning an object like blasthole drills are either from a US system called GPS or a Russian system called GLONASS i.e. Global Orbiting Navigation Satellite System. By using specialized components it is also possible to simultaneously use both the satellite and the constellations and get better accuracy.

The GPS system has 24 satellites in orbits at a distance 20200 km from center of the earth. Each of these satellite is capable of receiving and transmitting signals at different frequencies of radio waves for different communication channels. Barring exceptions, generally at least three satellites of GPS are visible from any point on the earth's surface. GLONASS visibility is even better than this. Ground control stations continuously monitor the position of the satellites at specific points with respect to the center of earth in the following manner:

- Each satellite always remains exactly at the same distance from the center of earth.
- Each satellite moves around the earth in a very specific direction at the same rotational speed as that of the earth
- From any given spot on the surface of earth a minimum of four to six satellites are visible so long as the spot is open to sky without any manmade or natural cover

With this, the position of each of the satellites remains unique and unchanged. In other words, if with respect to a point on the ground surface of the earth, a satellite is on a line that is at an angle $57^{\circ} 31' 27.819''$ above the horizon in a vertical direction, at an horizontal angle of $22^{\circ} 5' 7.331''$ measured from east in northerly direction and at a distance 22783.782 m from the point, it will always remain precisely at these angular parameters and distance with respect to that point.

Location recognition by GPS: If the signal sent from a blasthole drill to one satellite indicates that its distance from that satellite is X_1 , it means that the drill is somewhere on an imaginary surface of a sphere of radius X_1 with its center at that satellite. If at exactly the same time the distance of the drill from another satellite is determined to be X_2 , then the drill is somewhere on a circle that is formed by the intersection of two spherical surfaces, one with radius X_1 and the other with radius X_2 .

If at exactly the same time the distance of the drill from a third satellite is determined to be X_3 , then the drill is at one of the two points where the surface of the sphere of radius X_3 with its center at the third satellite intersects the circle cited in the paragraph above.

Which of the two points is the correct position of the drill can be determined in the same manner by using a distance measurement from a fourth satellite or even by knowing the position of a fixed tower in the vicinity of the drill because the actual position of the one of the two points will usually be at a very short or long distance from such tower.

Once the position of the drill can be precisely and unambiguously determined from each of the three or four satellites in the above manner, it can be transformed into latitude, longitude and altitude of the drill by using trigonometrical formulas because the position of each of the satellites is firmly fixed with respect to any point on the surface of the earth.

Movement of drill by using GPS: On a mining bench the position of each of the blastholes is precisely predefined. If such positions of the blastholes with their identifications are fed to the computer placed on the drill, it can be displayed on the screen of the computer. Similarly the position of the blasthole drill itself, as determined by GPS, can also be superimposed on the screen. Both these can be seen on a screen display shown in Figure 10.

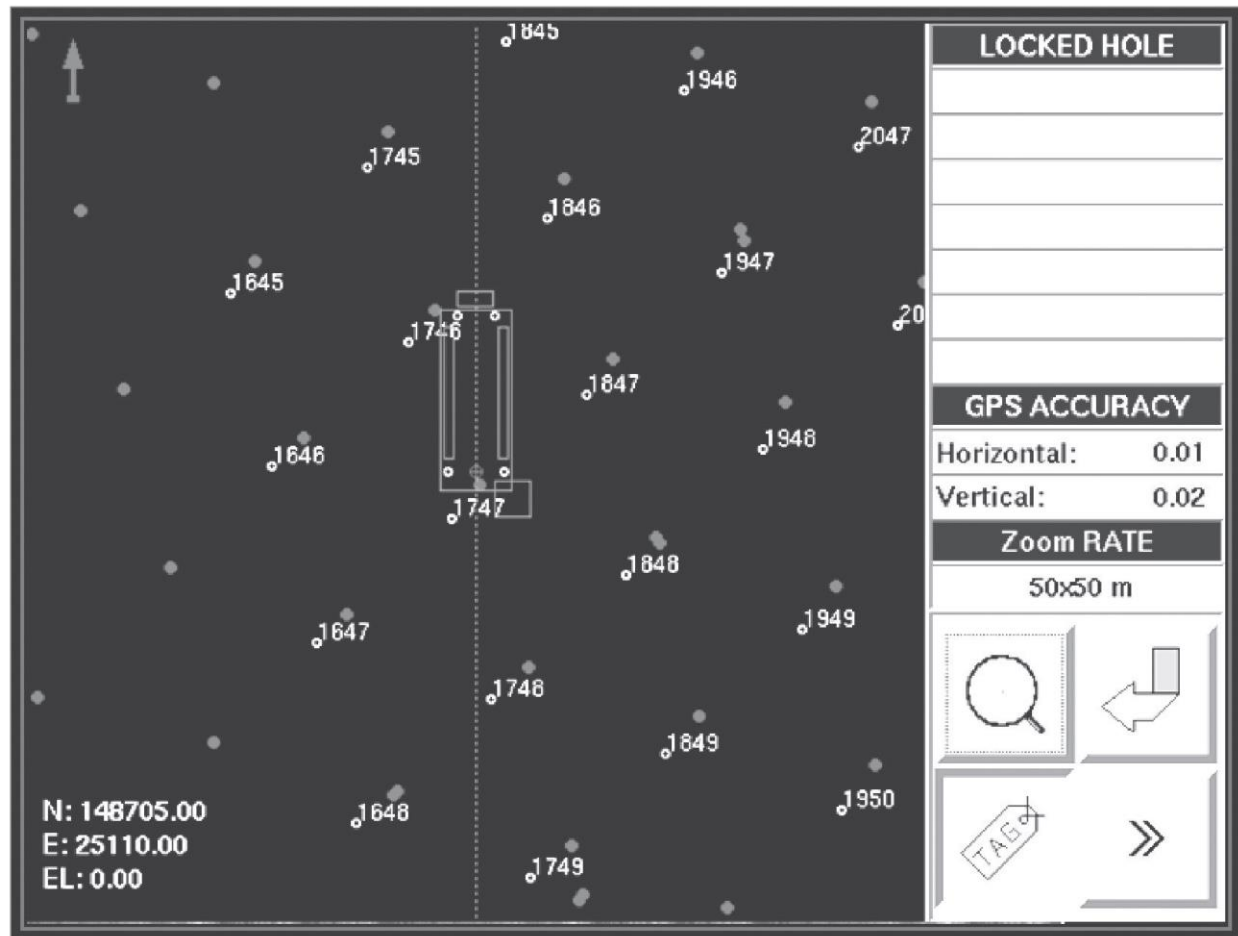


Figure 10: Blasthole drill navigation aided by GPS

The sequence of drill movements from position of one blasthole to the other is also well defined. Thus, a blasthole drill can be moved by the driller to the next blasthole position by looking at the screen in the operators cab rather than relying a flag mark made on the ground. As the drill moves the image of the drill on the screen also moves because the satellites are tracking the movement of the drill continuously. When the drill approaches the new blasthole, the screen image automatically gets magnified so the driller can position the drill more precisely. Finally when the drill has moved to the new blasthole within the tolerance parameters predefined for positioning, a signal is generated by the computer informing the driller that the drill has moved to the new blasthole precisely and he can start the next operation in the drilling cycle i.e. leveling the drill. Depending upon the type of equipment used, the accuracy possible with GPS can be as high as 5 cm meaning that the drill can be positioned within 5 cm of the precise position.

COMPUTERIZED DRILL SYSTEMS

Many computerized drill systems are available from many manufacturers. Components, specifications and accuracy of the digital drill recording system differs from manufacturer to manufacturer. A complete computerized drill system used on a blasthole drill comes with various degrees of sophistication. By and large the complete computerization of a rotary blasthole drill can comprise a drilling knowledge system, an automatic leveling system, an automatic drilling system and GPS positioning system. Further the complete

system can be integrated with an integrated mining system, which not only controls the blasthole drill operations but also the shovel and truck or dragline control system.

The main computer needed for running the programs of drill knowledge, drill automation, GPS and integrated mining system is placed in the operator's cab of the rotary blasthole drill and works in collaboration with the main computer usually located in the mine office. The position of the GPS components that go in the drill is shown in Figure 11.

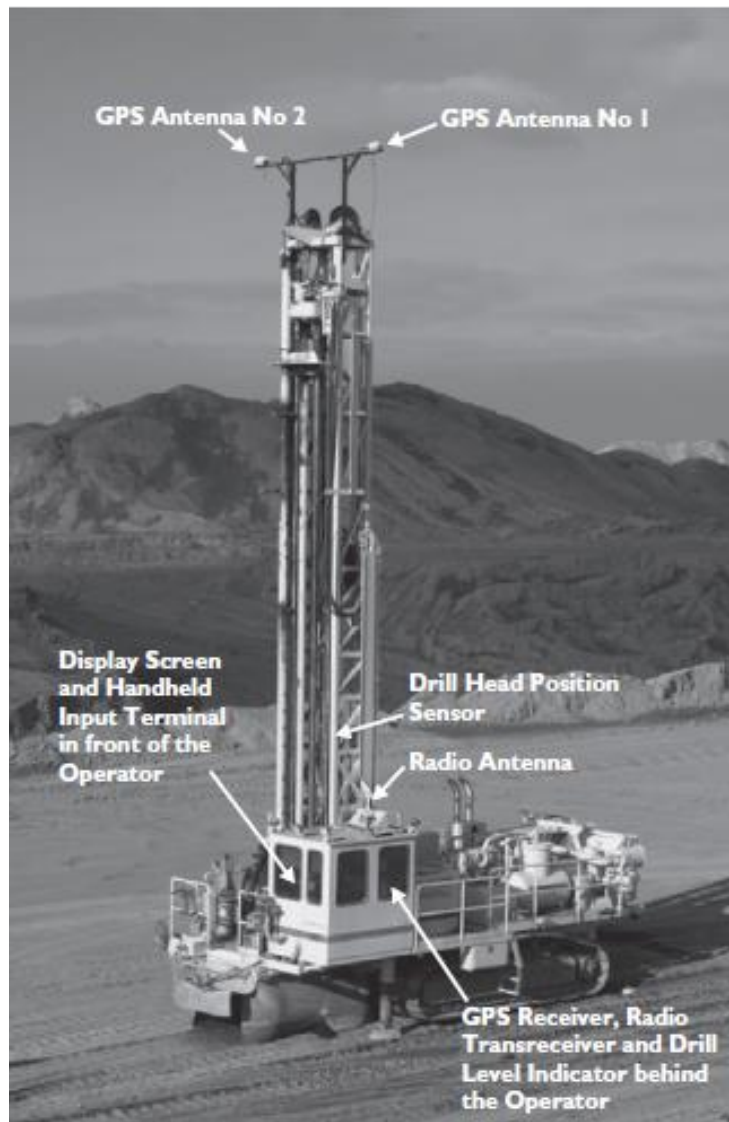


Figure 11: Position of GPS components in a blasthole drill

MEASUREMENTS WHILE DRILLING

Since the data required by a drilling knowledge system is acquired while the drilling operation is being carried out, it is often called 'Measurement While Drilling' and abbreviated as MWD. The MWD module can be incorporated in a blasthole drill to store the data measured while blastholes are being drilled. The MWD module can be incorporated in a blasthole drill to store the data measured while blastholes are being

drilled. A few parameters in the drilling knowledge system are basic in nature. They are either natural or are operator-controlled and are called independent parameters.

Independent parameters.

Time: This basic element is essential in most of the computations. In drilling knowledge systems, the accuracy of the clock need not be more than a hundredth of a second. Hence, usually the clock of the computer used in the system is sufficient. Measurement of time naturally includes date, as date is a designation given to a certain time interval of which second is the basic unit of measurement.

Depth: In drill knowledge systems, the depth is always known by measuring the travel of the drill head. Naturally, it is the length of the blasthole and not vertical depth. In most of the cases the depth measurement accuracy of 1 in 100 is sufficient. To actually measure the drilled movement, an optical encoder is fixed near the sheave, over which the chain or the rope that shifts the drill head travels. The optical encoder senses the rotational movement of the sheaves or shaft and converts it into the distance traveled by the drill head.

Rotary speed: The rotary speed of the drill string can be accurately measured by an optical encoder fixed on the drill head or by measuring the rotary speed of the hydraulic or electric motor used to drive the drill head and using the gear reduction ratio of the drill head in the calculations. Under most circumstances the accuracy of 1 in 200 is sufficient in rotary speed measurement.

Weight on the bit: Depending upon the system of exerting feed force on the drill head, every blasthole drill has a facility of measuring weight on the bit. In computerized systems these measurements can be collected and used for knowing the weight on the bit more precisely by adding the weight of the drill head and other drill string components in the drill string. Under most circumstances the accuracy of 1 in 1000 is sufficient in bit load measurement.

Torque: Torque is the force required for rotating the drill bit and the drill string. Torque is a good indicator of rate of penetration and hardness of rock.

Vibrations: Vibrations experienced by the drill string are eventually transmitted to the drill mast through the drill head. A vibration sensor mounted on the drill mast can easily give an idea of the very rough treatment experienced by the mast. A vibration sensor is required to measure both the amplitude and frequency of the vibrations in lateral, longitudinal as well as vertical directions. Vibration sensors are of particular importance from the viewpoint of automated drilling systems for the reason that vibrations reduce the fatigue life of the drill head and mast. Theoretically, in an extremely rare situation, the vibrational frequency of the drill string can match with that of the mast, and resonance can occur. In such an event the amplitude of the vibration increases to such an extent that it instantly causes heavy damage to the mast. Therefore, in automated drill systems the rotary speed and feed force are adjusted to reduce the amplitude and frequency of the vibration when these parameters start exceeding a set limit.

Compressed air pressure: This parameter is invariably measured and shown to the operator by a pressure gage in the operator's cab. For introducing compressed air pressure in a drilling knowledge system, a suitable pressure sensor is placed in the flow conduit.

Calculated parameters

Some parameters that have great relevance to the productivity and economics of rotary blasthole drilling and subsequent mining operations, have to be calculated from the independent parameters.

Rate of penetration: Rate of penetration is probably the most important parameter from the viewpoint of rotary blasthole drilling. Actually it is calculated by dividing the distance that the drill head travels while drilling is carried out, by the time spent for travelling the distance. The instantaneous penetration rate is $P_i = \Delta L_1 / \Delta T$. In most of the drilling knowledge systems the distance interval ΔL_1 can be set to a fixed distance and the average penetration rate for that small interval is calculated.

Blastability index: Blastability index is the most important parameter from the viewpoint of post-drilling operations such as blasting, loading, hauling, crushing etc. Unlike rate of penetration, blastability index cannot be directly calculated from the data acquired through the drilling knowledge system alone, but requires great calibration. Some diamond core drill holes have to be drilled at the worksite and the cores tested to ascertain the UCS of the formation. The UCS values so determined, and the presence of joints and cracks etc., are initially used for correlating the data acquired in drilling knowledge with a realistic blastability index. The methods and the mathematical formulation involved in establishing the correlation are often proprietary.

BLASTING IN SURFACE MINES

EXPLOSION PROCESS

Before discussing explosion process it becomes necessary to understand combustion, deflagration and detonation. During combustion actual burning always takes place within a small layer confined by two surfaces. This layer always moves to burn new material. Behind this layer the material is in burnt form and ahead of the layer the material is in unburnt form. The thickness of the burning layer depends upon the burning material itself and also on the physical extent of the burning material in continuum. Deflagration and detonation are the two distinct types of burning. When the speed of propagation of the burning layer is less than the speed of sound, the burning is called deflagration. In deflagration the burning layer advances by transferring heat energy to the adjacent layer in the unburnt portion. When the speed of propagation of the burning layer is in excess of 2000 m/s the burning is termed detonation. In detonation the burning layer advances by transferring energy through a shock wave. The speed of propagation of the burning front depends upon the intensity of heat with which the burning process has been started and how quickly, and also how much oxygen the burning process gets. Therefore, some explosives merely deflagrate when brought in contact with a common match-stick fire but detonate and explode if brought in contact with a spark that has very high intensity of heat. Usually all materials that detonate are called explosives. Detonation of all the explosives results in a very large volume of gases and a very large quantum of heat.

Most of the explosives are chemicals formed by carbon, hydrogen, oxygen and nitrogen. When they detonate, they mainly generate common and harmless chemical compounds such as H₂O (Water), CO₂ (Carbon Dioxide), N₂ (Nitrogen) with a very small quantity of poisonous gases such as NO (Nitrous Oxide), CO (Carbon Monoxide), NH₃ (Ammonia), CH₄ (Methane).

In the case of explosives the velocity of propagation of the burning layer is called the velocity of detonation (Figure 12). The frontal surface of the burning layer, more commonly called the detonation zone, is termed the shock front and the rear surface is called the Chapman-Jouguet surface.

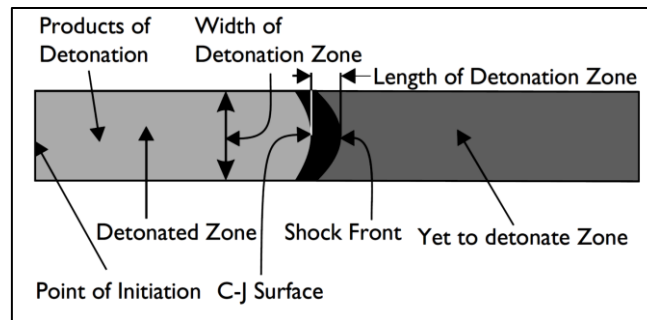


Figure 12: Propagation of detonation

The length of the detonation zone depends upon the components of the explosive, particle size, density and confinement of the explosive. The width of the detonation zone is proportional to the length of detonation zone. For every explosive there is a certain blasthole diameter called the ‘critical diameter’. If the particular explosive, filled in a blasthole having a diameter smaller than the critical diameter is detonated, there is a likelihood that detonation of the explosive will be incomplete and blasting will be ineffective. Thus for effective use of the explosive, the actual blasthole diameter must be more than the critical diameter.

BLAST FRAGMENTATION MECHANISM

The many disintegration modes that contribute to the process of rock fragmentation are as under:

1. Crushing of Rock
2. Radial Cracking
3. Circumferential Cracking
4. Spalling
5. Radial Push by Gases
6. Cracking by Flexure
7. Collision of Fragments
8. Separation of Beds

Crushing of rock: Explosive filled in a blasthole is called the charge. The process of systematically filling charge in a blasthole is called charging. Initially when the explosive charge is detonated by a spark at the bottom of the blasthole the detonation zone starts moving up the blasthole at a speed of 2500 to 7000 m/s. The energy released and the gases generated in the detonation zone exert an extremely high pressure against the walls of the hole. Simultaneously the heat generated in the detonation gives rise to very high temperatures. In most cases the pressure ranges between 7 to 10 GPa and temperature ranges between 2500 to 4500°C. Such high temperature and pressure act together to pulverize and crumble a thick cylindrical layer around the blasthole wall. This is called the crushed zone. The thickness of the crushed zone depends upon the magnitude of the detonation pressure, heat of detonation, and the strength and porosity of the rock.

All this culminates in an increase in the volume of the blasthole. If VO is the original volume of the blasthole and VE is its volume after explosion, the increase in ratio VE/VO with time is as shown in Figure 13. This ratio is of the order of 2 to 4 for strong rocks and of the order of 10 for weak and porous rocks. As much as 30% of the energy generated in the detonation zone is spent in the crushing of rock.

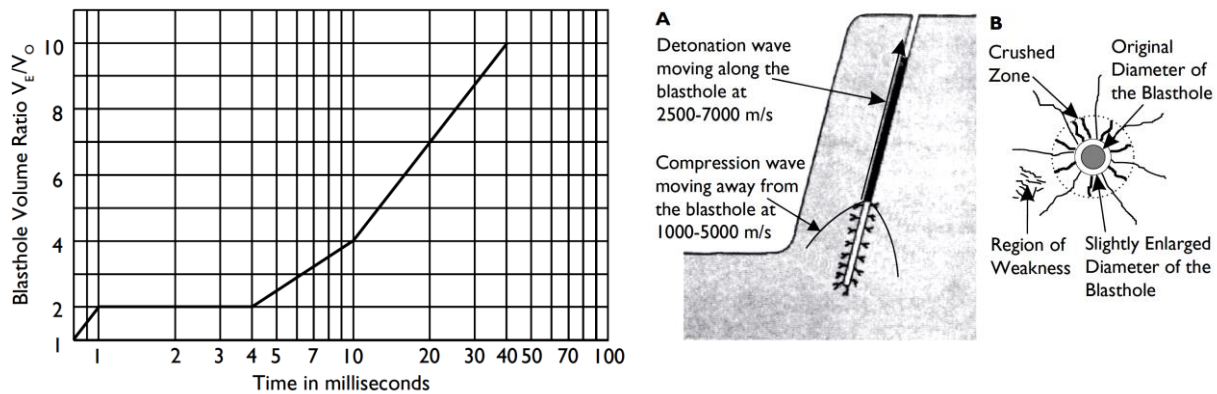


Figure 13: Increase in blasthole volume after detonation and Initiation of rock fracture by blasting

Radial cracking: A compression wave is also created at the detonation zone. It starts moving away from the blasthole in all directions at the velocity of a sound wave in that medium, as shown in Figure 13 A. All rocks are formed by minerals of different composition. Therefore, planes of weakness are left during the process of formation of a rock mass. The cracks, which are in a direction tangential to the wave propagation, remain nearly unaffected by the compression wave. The compression wave generates shear stress along the radial planes of weakness because the response of different crystals to compressive stress differs. The number of cracks in the region adjacent to the walls of a blasthole is higher, but at longer distances from the blasthole the formation of cracks is lesser as shown in Figure 13 B. This happens because there is a significant reduction in the energy contained in the compression stress wave. As the rock mass on the rear side of the blasthole is in horizontal confinement, there is no significant radial cracking on the rear side of the blasthole.

Circumferential cracking: Within a very short period the compression wave reaches the free surface at the bench face and gets reflected as a tension wave. The speed at which the tension wave travels backward is about half of the speed of the compression wave, i.e. 500 to 2500 m/s. The lower speed of the tension wave is due to the fact that a significant part of the energy contained in the compression wave is lost as it moves to the bench face. The quantum of energy lost depends upon many factors such as composition of the rock mass, the distance between the blasthole and the free surface nearest to the blasthole at the time of detonation (i.e. the burden), the distribution of pre-existing cracks in the rock mass, the porosity of the rock mass etc. Tension waves are very important from the viewpoint of blasting because they generate circumferential cracks in the rock mass as shown in Figure 14 A. Without such circumferential cracks there cannot be any significant fragmentation in the rock mass. The portion between the blasthole and the bench face is heavily cracked by the tension wave as shown in Figure 14 B. The reason for large scale circumferential cracking in the rock mass by the tension wave is that all the rocks have very low tensile strength. Since there is no free surface on the rear side of the blasthole there is no circumferential cracking on the rear side.

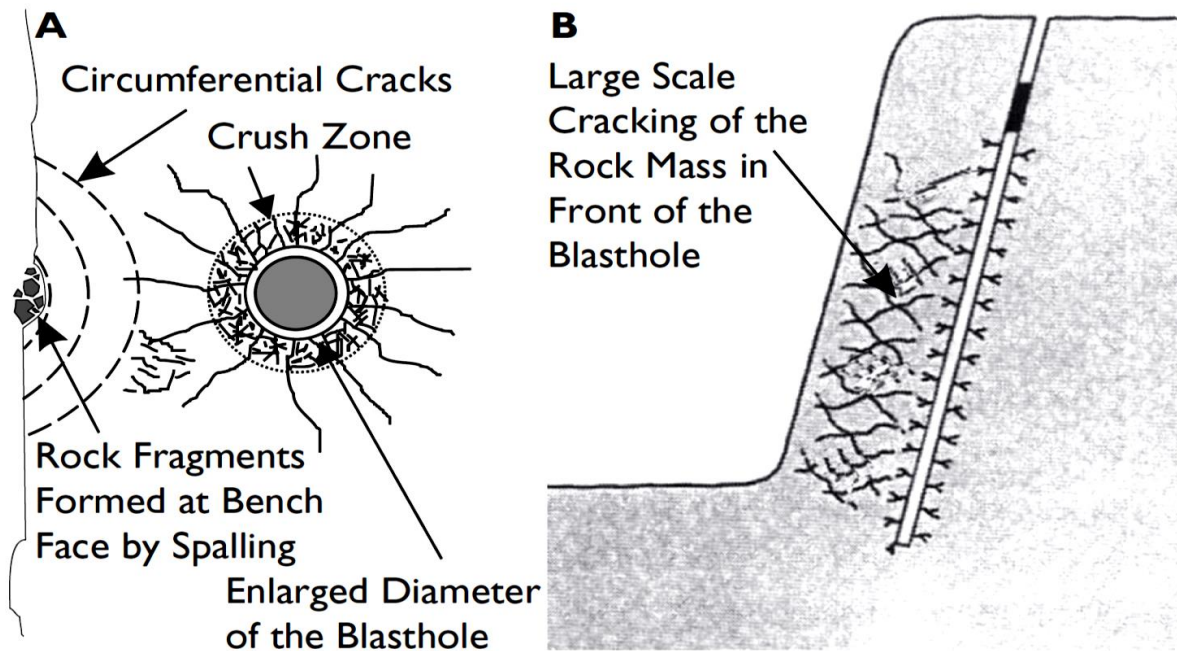


Figure 14: Rock mass cracking by tension wave

Spalling: When a compression wave reaches the bench face and reflects as a tension wave, in the very initial stage circumferential radial cracks develop near the bench face. Since there is no confinement of the rock fragments on one side of the bench, they separate from the bench face and fall down. This fragmentation is called spalling.

Radial push by gases: After the radial and circumferential cracking the rock mass on the front side of the blasthole is already loosened, but remains without significant movement. Gases formed in detonation and held at exceedingly high pressure start flowing to the circumferential cracks through the radial cracks and exert outward pressure. This causes throw of the rock fragments in an outward direction i.e. away from the bench face as shown in Figure 15.

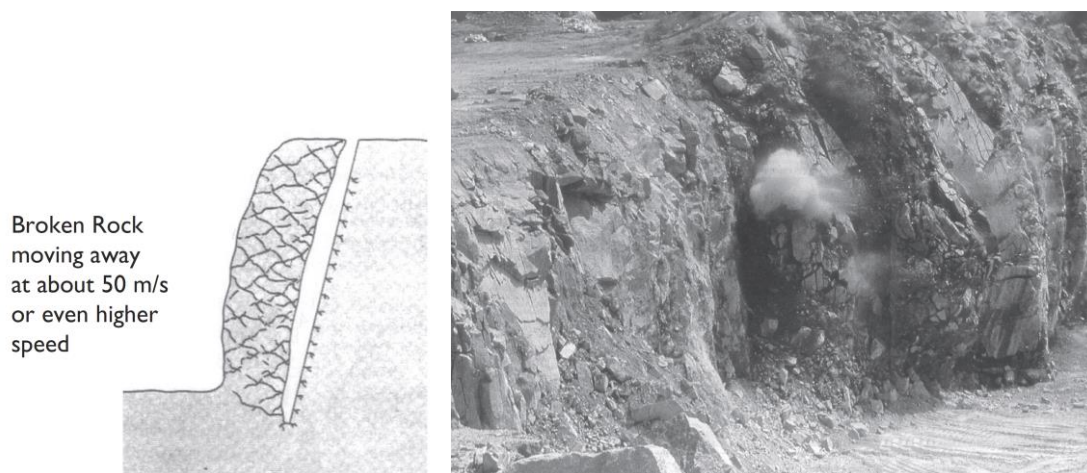


Figure 15: Throw of rock fragments

Cracking by flexure: In a blasthole the detonation process is almost invariably started at the bottom of the blasthole. In the initial stages gases form at the bottom and exert upward pressure. As the detonation layer moves upward, the gases formed in the upper region also add to this pressure. Thus, the pressure in the middle of the blasthole length is higher than the pressure at the bottom. This differential pressure creates a bending moment on the burden wall as shown in Figure 15 and as is evident from the photograph in Figure 15. With such flexure given to the burden wall, horizontal cracks form in the wall on the outer side i.e. near the bench face.

Collision of fragments: When fragments formed by blasting are thrown away from the rock mass in burden, they collide with each other. Such collisions result in further fragmentation of the pieces of rock.

Separation of beds: If the rock mass consists of different layers, as usually happens in sedimentary rock formations, the response of different layers to the action of the compression wave is different. For this reason, in many cases the beds on two sides of the bedding plane get separated by cracks formed at the bedding plane. Such cracks invariably add to the process of fragmentation by blasting. The speed at which the fragments are thrown away depends upon the pressure and rock mass characteristics.

All the modes of rock fragmentation described above take place in such a short time interval that it is very difficult to give any sequence.

EXPLOSIVES

Many materials burn. Whenever they burn, there is a layer in which the burning action is in progress. On one side of the layer the material is in burnt form. The burning layer propagates towards the other side where the material is yet to be burned. The speed of propagation of the front surface of this layer differs from material to material. Based on the speed of propagation, five classes of burnable materials viz. combustible, inflammable, highly inflammable, low explosives and high explosives are usually recognized. Table 3 clarifies the differences in these classes.

Table 3: Classes of burnable materials

Class	Classification norm (m/s)	Typical material
Combustible	Less than 0.01	Wood, Coal
Inflammable	Between 0.01 to 1	Kerosene, Paraffin
Highly Inflammable	Between 1 to 700	Petrol, Sulfur, Magnesium
Low Explosive	Between 700 to 2000	Black Powder
High Explosive	More than 2000	Nitroglycerin, TNT

For a chemical to be considered as an explosive, it must exhibit the following characteristics.

1. Rapid expansion
2. Quick liberation of heat
3. Fast reaction
4. Need to initiate the chain of reactions

Rapid expansion can occur when a material burns to evolve gases. Thus, an explosive must be largely composed of gas-forming elements viz. carbon, hydrogen, oxygen and nitrogen. Further, for the rate of burning to be high, either the particles must have sufficient oxygen content so the chemical action readily

gives oxygen, or the size of particles of the material must be very fine so that the atmospheric oxygen is available in close vicinity of a very large surface area.

Coal burns slowly in atmospheric air. If it is finely powdered it can burn in air very rapidly. If it is mixed with liquid oxygen it can burn violently and cause an explosion.

Quick liberation of heat in the chemical reaction can occur depending upon the composition of the material. The gases formed in the chemical reaction expand and have devastating effect only when quickly-released intense heat is available.

A kilogram of coal has five times the calorific value of one kilogram of nitroglycerin. However, detonation of nitroglycerin causes an explosion and gives much higher shattering effect than the slow burning of coal.

Rapid expansion of the material and quick liberation of gases can take place only when the chemical reaction takes place very rapidly.

For a chemical to be considered as an explosive, the explosion must occur only when an instance designed by a human occurs. Without such a characteristic the explosive cannot be very useful.

The process of very rapid burning of the low explosives is called deflagration, whereas the process of lightning-fast burning of high explosives is termed detonation.

Low explosives do cause a heavy push or powerful lift of the material that surrounds them but do not cause a shatter. Often, low explosives are mixtures of a combustible substance and an oxidant that decomposes rapidly to give a high quantity of oxygen for the quick burning.

High explosive materials decompose very rapidly through detonation under certain circumstances to evolve a huge volume of gases, extremely high quantity of heat, rapidly traveling shock waves in atmospheric gases as well as the ground rock mass, light, deafening noise and hazardous fly rock. The build-up of pressure on the walls of a blasthole upon blasting low and high explosives is shown in Figure 16 further is shows the classification of high explosives.

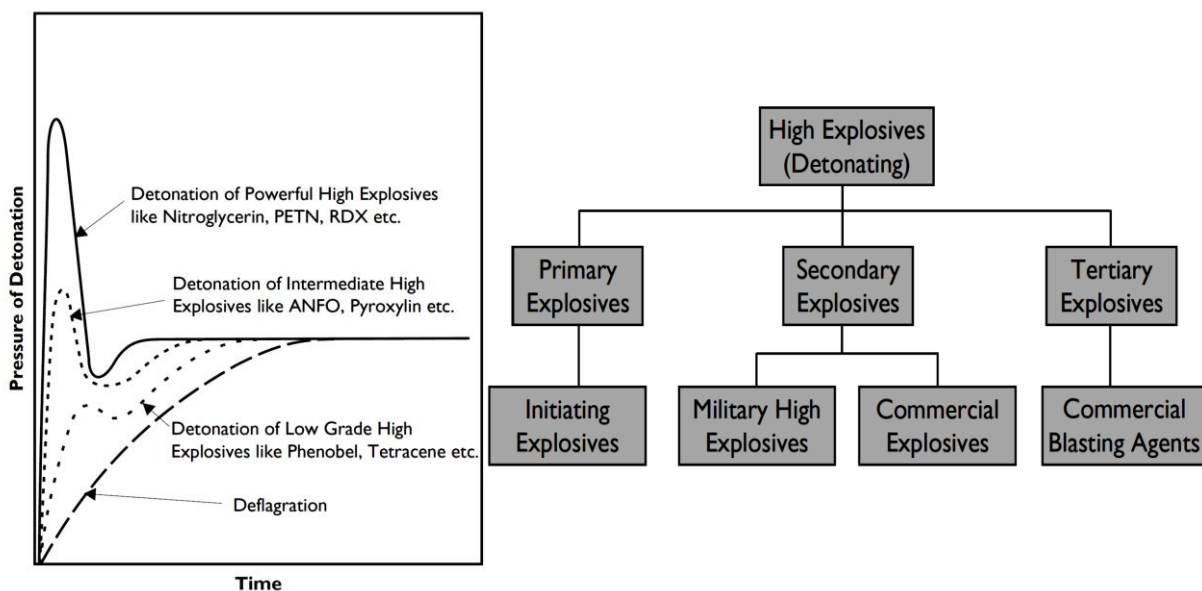


Figure 16: Build-up of pressure on the walls of blasthole and classification of high explosives

Primary explosives are very easily influenced by heat, friction or shock to cause detonation. Commonly used primary explosives are mercury fulminate, lead styphnate and lead azide. These are also termed as initiating explosives. PETN is a benchmark explosive. All explosives more sensitive than PETN are primary explosives.

Secondary explosives are relatively insensitive to heat, shock or friction. They are also called base explosives. Since they are not easily influenced by heat, friction or shock, they can be cautiously used in large quantities during warfare as well as rock blasting.

Some secondary explosives may burn when exposed to heat or flame in small, unconfined quantities, but detonation can occur. Secondary explosives usually need a small device containing a very small quantity of primary explosive for their detonation. Most of the secondary explosives are chemical compounds rather than mixtures. Explosives, such as dynamite, TNT, PETN, RDX, HDX etc., are mainly used for warfare activities. Some of these, particularly dynamite, were used for rock blasting in the early days but now they are being replaced by tertiary explosives.

Tertiary explosives are those which are very insensitive to shocks. They cannot be reliably detonated by a limited quantity of primary explosive. Therefore, the detonation device used for them contains a small quantity of secondary explosives. Explosives in a confined state that cannot be detonated by means of test blasting cap no. 8 are classified as tertiary explosives.

Tertiary explosives are also called blasting agents. Tertiary explosives contain primarily an inorganic nitrate and a carbonaceous fuel. For certain purposes they also contain non-explosive substances like powdered aluminum. The most commonly used tertiary explosive is ANFO. Almost 80% of mining blasts are made by using it. ANFO is actually a mixture of ammonium nitrate and fuel oil. Slurry or “wet bag” explosives are also tertiary explosives.

Properties of explosives with their meaning is given in table 4.

Table 4: Properties of explosives

Explosive property	Meaning
Velocity of Detonation	Velocity in m/s at which the shock front of the detonation layer travels within the column of explosive.
Detonation Pressure	Pressure developed by detonation of the explosive in the detonation zone. It is usually measured in GPa.
Blasthole Pressure	Pressure exerted on the wall of the blasthole immediately after the detonation.
Strength	Total amount of energy released by the explosion in MJ for each kg of explosive. This includes the energy released in the form of heat as well as the pressure exerted by the gases generated in detonation.
Heat of Explosion	Total amount of heat released by the detonation in kcal for each kg of explosive.
Specific Gas Volume	Amount of gas generated by detonation of one kg of explosive under normal conditions.

Explosive property	Meaning
Sensitivity	Possibility of causing detonation by such means as friction, pressure, heat etc.
Transport and Handling Safety	How easily the explosive can be handled and transported through different modes of transport.
Brisance Value	Brisance value indicates the shattering effect of the explosive.
Charging Density	Weight of the explosive in kg, contained in each liter volume of the blasthole.
Toxic Fumes	Volume of poisonous gases generated in terms of liters per kg of explosive detonated.
Water Resistance	Whether the properties of explosive remain unchanged by mixing the explosive with water.
Hygroscopicity	Hygroscopicity is a measure of water-absorbing capacity of an explosive
Minimum Hole Diameter	What is the smallest diameter of blasthole in which the explosive can be charged and detonated to get the desired explosion effect.
Storage Life	How long the explosive can be stored in originally packed and unpacked condition without a change in its properties.
Volatility	How much is the volatility of the explosive
Material Coexistence	This is the ability of the explosive to coexist with other materials

BRIEF HISTORY OF EXPLOSIVES

Some chemicals, or mixtures of them, that can be considered as explosives by current definitions of the word, may have been known to mankind long before the Christian era, but there are no specific written references to them. Armaments with explosive-type devastating power are said to have been used in the Mahabharata War supposed to have been fought as early as 5000 BC. Black powder was known to the Chinese from the eighth century but was used only for entertaining the public.

Table 5: Developments in Explosives

Year (CE)	Developments in Explosives
1200	Arabian writer named Abd Allah mentioned that saltpeter (Potassium Nitrate) is the main ingredient of black powder.
1242	English friar Roger Bacon published the formula for making gunpowder through his book 'De Mirabili Potestate Artis et Nature'. It was in coded language. He also gave details of the destructive power of the powder.

Year (CE)	Developments in Explosives
1380	German Franciscan monk, by the name of Berthold Schwarts, developed gun powder i.e. black powder. He also used it in guns.
1617	Use of black powder for the purpose of rock blasting was first proposed by Martin Weigel, a mining superintendent at Frieberg, Germany.
1623	Casper Windt is said to have actually used black powder for rock blasting at a German town Schemitz
1670	German miners revealed the secret of using gun powder for the purpose of rock blasting in Cornwall England.
1675	A factory for making black powder was established in what is now Milton, Massachusetts, USA.
1696	The first instance of the use of black powder for blasting rock in a road-widening project occurred in Switzerland.
1745	Doctor Watson invented the technique of exploding black powder by electric spark.
1750	Benjamin Franklin invented a cartridge by compressing black powder in a cylindrical shell with a electric spark generating device in it to cause detonation.
1773	Black powder was used for mine blasting for the first time in a copper mine in Connecticut in what is now the USA.
1822	The first hot wire detonator was produced by Dr. Robert Hare. Using one strand separated out of a multi strand wire as the hot bridge wire, this blasting cap ignited a pyrotechnic mixture (likely to be potassium chlorate/arsenic/sulfur) and then a charge of tamped black powder.
1830	Moses Shaw of New York patented a device for electric firing of black powder
1831	William Bickford of Cornwall, England invented a safety fuse.
1832	Dr. Robert Hare developed the electric blasting cap.
1846	An Italian chemist, Ascanio Sobrero, discovered nitroglycerin. While doing so he was severely injured by the explosion. He therefore never revealed the formulation.
1863	Wilbrand invented TNT i.e. Trinitrotoluene. In the same year Alfred Noble developed a safer variety of nitroglycerin for use in rock blasting. It could not be exploded by a normal fuse.
1864	Alfred Noble developed the first detonating cap. This became essential for detonating the safer variety of nitroglycerin.
1866.	Alfred Noble invented dynamite by mixing kieselguhr (kind of soft rock containing the remains of diatoms) with nitroglycerin.

Year (CE)	Developments in Explosives
1870	A plant was set up by the Giant Powder Company for making dynamite at San Francisco, California, USA.
1875	Alfred Noble patented blasting gelatin (an explosive material consisting of collodion-cotton, a type of nitrocellulose or gun cotton, dissolved in either nitroglycerine or nitroglycol and mixed with wood pulp and saltpeter). In the same year Perry, Gardiner and Smith independently developed and marketed caps which combined the hot wire detonator with mercury fulminate explosive. These were the first generally modern type blasting caps.
1884	Large scale use of ammonium nitrate, commonly abbreviated as AN, as an additive to dynamite began.
1902	Detonating cord was introduced in Europe
1913	Explosive was used by M. Kinley for extinguishing a wildly burning oilwell in California. It worked on the principle that the sudden change in pressure across the shockwave, and then the impulse of the airflow behind it pushed the flame straight off the fuel source. As soon as the flame didn't have access to fuel anymore, it stopped burning.
1917	L. Mintrop, a German scientist, invented the seismograph
1921	A devastating explosion of ammonium nitrate took place in Oppau, Germany. This explosion brought the explosive nature of ammonium nitrate to the notice of scientists.
mid 1920s	Liquid oxygen-based explosives made their appearance in the USA.
1935	a machine for measuring blast vibrations was developed by the US Geological Survey
mid 1950s	High speed photography was introduced in mining for analyzing a blast.
1955	Bob Akre developed "Akremite" (a mixture of prilled ammonium nitrate, fuel oil and dynamite)
1956	Mixture of ammonium nitrate and fuel oil was used for the first time in US Steel Corp.'s Oliver Mining Division for rock blasting. This two-component explosive later got the acronym ANFO. In the same year Mel Cook introduced slurry explosives in mining practice.
1957	A specialized vehicle which could mix the two components of ANFO in precise proportions inside the blasthole was developed in Canada. By the end of the 1950s prilled AN and fuel mixture became so popular that it shunted out dynamite
1967	Shock tube type non-electric detonators were introduced in Sweden

Year (CE)	Developments in Explosives
1969	Emulsion explosives first appeared. Emulsion Explosives are intimate mixtures of oxidizers and carbonaceous fuels made as water in oil type of Emulsions.
1972	Computer modeling of blast design was developed by Land & Favreau
early 1970s	Electronic recording seismographs were introduced by Dallas Instruments
1973	Electronic sequential blasting was introduced at the first Kentucky Blaster's Conference
1974	Shock tube type non-electronic delay detonators were introduced
1974	The International Society of Explosives Engineers was formed in Pittsburgh, Pennsylvania, USA.
1977	Glass "bubbles" were introduced in mining blast practice
1980s	Digital sampling seismographs were developed. In this decade the earliest computer programs for improving blast timing patterns and blast plans were introduced in mining practice.
1988	The technique of laser profiling for blast design and analysis, that was developed in Britain, was introduced in USA
late 1980s	Electronic delay detonators were introduced.
1992	Nearly 700 oil well fires in Kuwait after the Gulf War, were put out by means of explosives.

TYPES OF EXPLOSIVES IN A BLASTHOLE

A blasthole may contain two to four of the following entities of explosives.

1. Detonator i.e. Blasting Cap
2. Primary Explosive
3. Booster Explosive
4. Main Explosive

A blasthole charged with them is shown in Figure 17. Other Blasting consumables are shown in Figure 18.

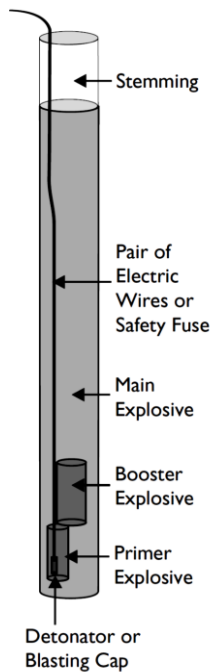


Figure 17: Charged Blasthole **Figure 18: Other Blasting Consumables**

Detonator (Blasting cap): A detonator is the device that starts the process of detonation in a blasthole. It is made from a very sensitive explosive such as lead azide or mercury fulminate (Figure 19). Initial energy delivery to the detonator is by means of a spark/heat generated through two electric wires or fire propagated up to it through a safety fuse. When a detonator is detonated by means of a safety fuse it is called a blasting cap.

Modern explosives used in mines are very safe. In other words they do not detonate unless a very high quantum of energy is imparted to them. This is the reason for the need of a detonator.

Detonation in a blasthole is always started from the bottom of the blasthole.

Hence, the detonator is always placed in the bottom of the blasthole after connecting it to a safety fuse or electric wires.

A detonator always consists of explosives that are sensitive to heat.

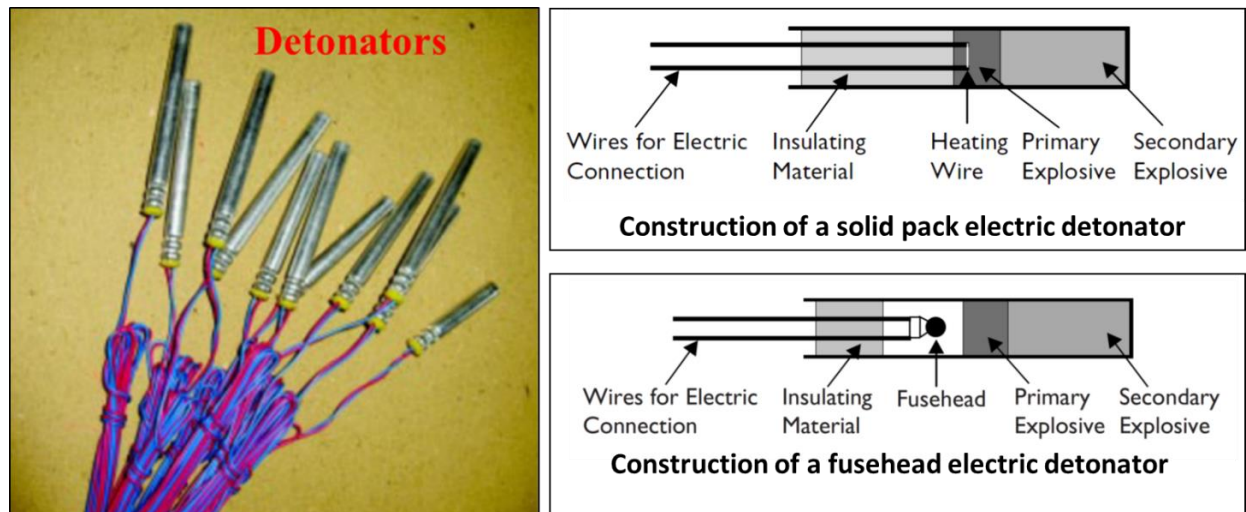


Figure 19: Detonators used for Blasting

Non-electric detonators are also called plain detonators. They are fired by detonating cord instead of electricity. A non-electric detonator consists of a plastic shell filled with primary explosive, secondary explosive and a delay if applicable, and a certain length of detonating cord. The construction of a non-electric detonator is shown in Figure 20.

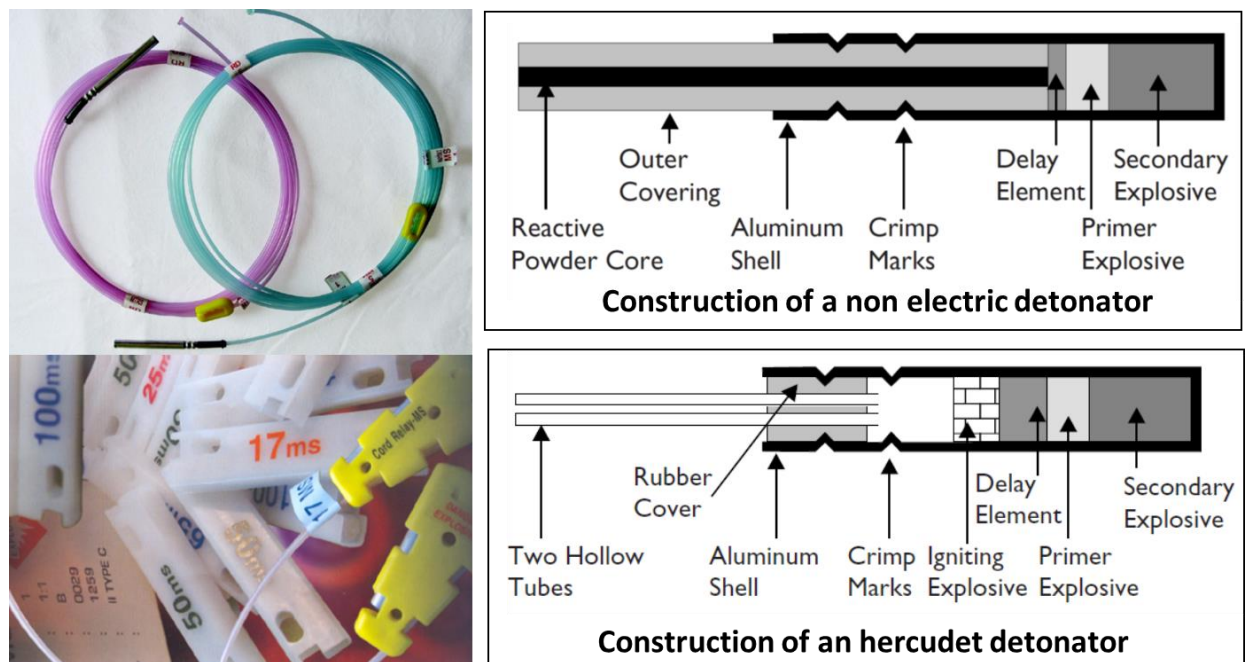


Figure 20: Non electric Detonator

In hercudet system the transmission tube is a simple plastic hollow tube, about 4 mm outside and 2 mm inside diameter. It does not contain any explosive or metal. By means of special straight or elbow connectors the tube forms an airtight circuit, much similar to household piping. Hercudet detonators are a part of such a circuit. When the circuit is completed it is first checked for air-tightness by pumping compressed air into it. Once it is confirmed that the circuit is airtight for pressure up to about 300 kPa, gaseous oxidizer and

gaseous fuel mixture is fed into the circuit. When the circuit is completely filled with the mixture the gas is ignited. The fire travels throughout the circuit at a speed of about 300 m/s to detonate all the detonators in the circuit.

Electronic detonators: As the microelectronic circuits became available easily and cheaply, electronic initiating systems were made by using them. Like electric initiating systems, these systems also use a detonator and electric wires but the detonator in an electronic system works on the basis of digital signals emanating from the initiation device located at a long distance on the ground rather than by electric current. A typical electronic detonator looks like the one shown in Figure 21.

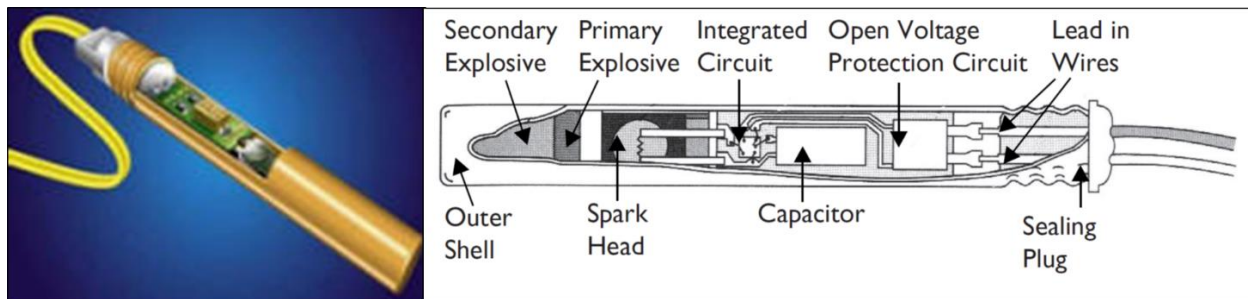


Figure 21: Construction of an electronic detonator

Primary explosive: Many explosives used in large surface mines are very insensitive. As they do not get effectively detonated even by using blasting cap no. 8, they are called cap insensitive. They need more powerful explosive to start the detonation of the main explosive in the blasthole. Primary explosives are made for this purpose. Primary explosives come in the form of a cylindrical-shaped cartridge and contain a detonator embedded within them. The explosives used for making primer are PETN, TNT, RDX, Tetryl, etc.

Booster: A booster is also made by using PETN, TNT, RDX, Tetryl, Picrite etc. It is used anywhere in the blasthole where there is a need for generating higher energy. Since both these are cap sensitive, they easily detonate by the detonation wave traveling in the main explosive. Therefore, they do not need any embedded detonator. The most common place for a booster is at the bottom of the blasthole where the burden is usually more than in the rest of the blasthole. At this place much higher energy is required to be generated so as to ensure that no stumps are formed and the rock mass in the wider burden gets well fragmented.

Main explosive: The main explosives used in mining blasts are usually multi-component type. They contain fuel, oxidizer and in many cases a sensitizer as well. Fuel is meant to burn and generate heat. The oxidizer ensures proper burning of the fuel without formation of poisonous gases. It also accelerates the process of burning. Where high energy output is needed a sensitizer is added to the explosive mixture. The fuel, oxidizer and sensitizer combination of Dynamite, ANFO, Slurry and Emulsions is given in Table 6.

Table 6: Composition of main explosives used in mine blasts

Explosive type	Oxidizer	Fuel	Sensitizer
Dynamite	Solid Nitrate Salts	Solid Carbonaceous Oil	Liquid Nitroglycerin, Voids, Bubbles
ANFO	Solid Ammonium Nitrate	Liquid Diesel Oil	Voids, Friction

Slurry	Solid or Liquid Salt Solutions of Nitrates	Solid or Liquid Carbonaceous Fuel, Aluminum	Solid or Liquid TNT, Fine Aluminum, Bubbles
Emulsion	Liquid Nitrate Salt	Liquid Carbonaceous Oils	Bubbles

DRILLING PATTERNS

On the surface of a bench blastholes are drilled in one of the three patterns, viz. square, rectangular, and triangular.

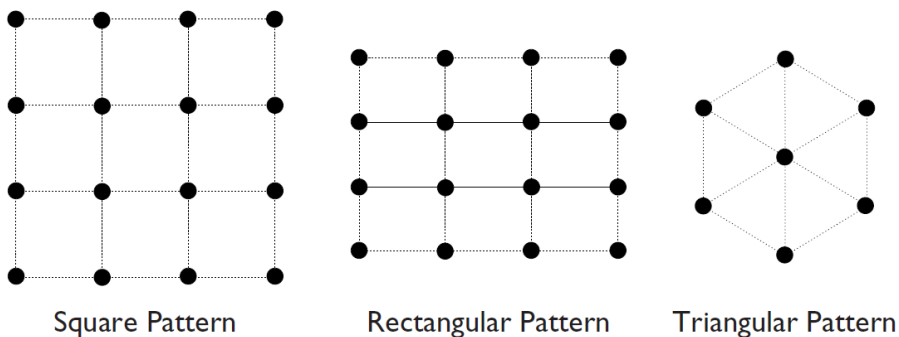


Figure 22: Blasthole layout patterns

The diameter of the blasthole is selected according to the thumb rule as tabulated below:

Blasthole dia. Mm	Bench heights in m																
	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
152.4																	
158.8																	
171.5																	
187.3																	
200.0																	
228.6																	
250.8																	
259.9																	
279.4																	
311.2																	
349.3																	
381.0																	
406.4																	

FIRING PATTERNS

In large surface mines five commonly adopted firing patterns are used. They are explained below.

1. **“V” Pattern** : The fragments of rock are in the form of a tall heap. Material in such a heap can be easily lifted by shovels and loaded into the dumpers. Since rock fragments are thrown towards each other in a 90° angle they collide with each other and further reduce in size.
2. **Echelon Delay Pattern**: It is particularly suitable for mine benches which have two free faces. This firing pattern gives a muck pile that is not flat but well spread on the bench floor. Such muck piles are suitable for loading by wheel loaders.
3. **Flat Face Pattern**: The flat face firing pattern is adopted when the number of rows in the blast is five or less. The drilling pattern of the blastholes is usually staggered or triangular because with such a drilling pattern the fragmentation is within acceptable limits. When the blast is meant for getting large stone pieces to be used in a dam or breakwater, a square or rectangular drilling pattern is better suited.
4. **Channel Pattern**: Sometimes mining activities have to start from the side of a hill. In such case there is no bench and so no bench top or bench floor. A classic excavation of this type of operation is a channel cut for some distance, made before the entry into a large tunnel. The blasted material must be made to accumulate in the blasted area where shovels can approach from one side and the dumpers also approach from the same side. For the purpose of loading muck into the dumper the shovel must slew through a 180° angle rather than the usual 100° to 120°. Under such situations the most suitable firing pattern is the channel delay pattern.

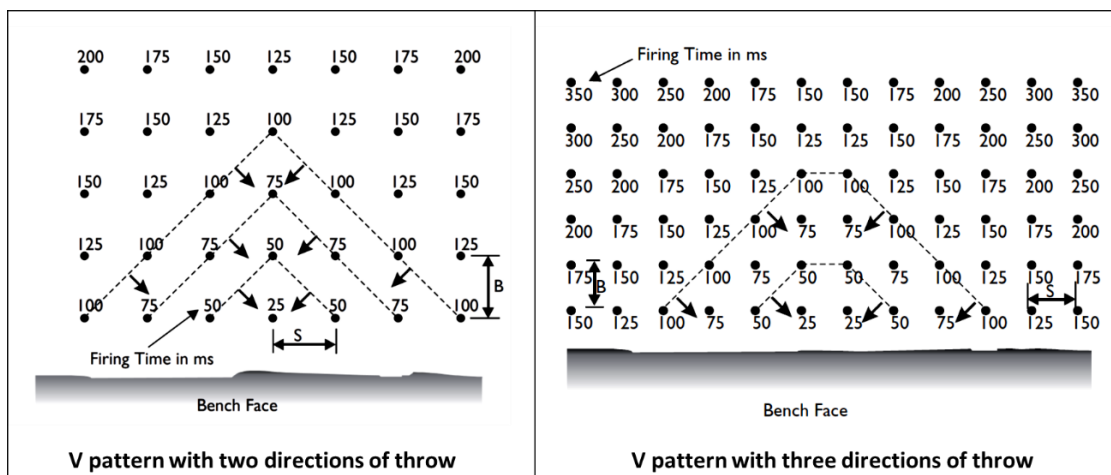


Figure 23: “V” Pattern of firing

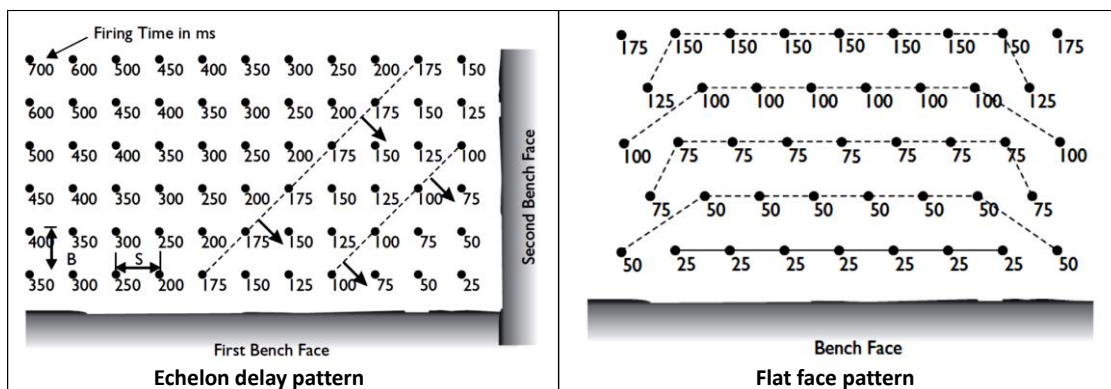


Figure 24: Echelon and Flat face delay patterns

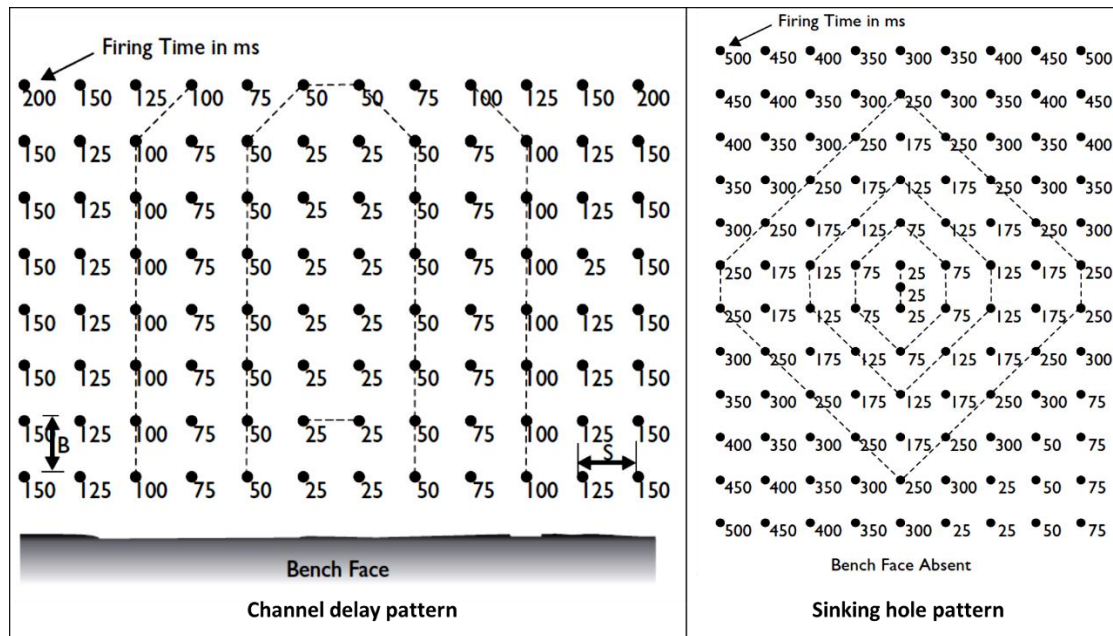


Figure 25: Channel and Sinking hole delay patterns

5. **Sinking Hole Pattern:** Many mines, particularly coal mines, are located on horizontal planes. To start a mine in such terrain, initially a huge excavation, much like the excavation for a foundation of a skyscraper, is started by blasting vertical holes. Blast starts with detonation of the explosives in centrally located blastholes. Later blastholes surrounding these holes, but which lie on the periphery of the diamond shaped area, are detonated with the same delay. In this manner the blast spreads outward. In such a blast the fragments do not move in a horizontal direction because there is no space for such movement. Naturally, movement of rock fragments in the upward direction is on a much larger scale. Ground vibrations and flyrock hazard of this type of blast is much higher than in other types of blasts with an equal quantity of explosive. To reduce both these hazards the delay intervals are kept large.

BLASTHOLES IN A MINE BENCH

A blasthole drill stands on a mine bench top and drills blastholes in it. Blastholes drilled in the rock mass appear as shown in Figure 26. In the early days of blasthole drilling, when churn drills were used for the purpose, the blastholes could only be drilled in a vertical direction. This trend continued during the dawn of rotary blasthole drilling as well. Now, inclined blastholes as shown in the Figure 26(b), are also drilled in many mines. Inclined blastholes often prove more appropriate from the viewpoint of reduction in cost. Ideally the angle of inclination of blastholes is 45°, but practical limitations of rotary drills limit the angle of inclination to 30°. On certain occasions, in addition to the main blastholes, some additional vertical or inclined blastholes, like those shown in Figure 26(c), are also required for the appropriate fragmentation of rock mass. These blastholes are of half or even quarter depth depending upon the characteristics of the rock mass. Such blastholes are referred to as satellite blastholes. Similarly, situations also arise when additional horizontal blastholes like those shown in Figure 26(d) become necessary. They are termed snake holes.

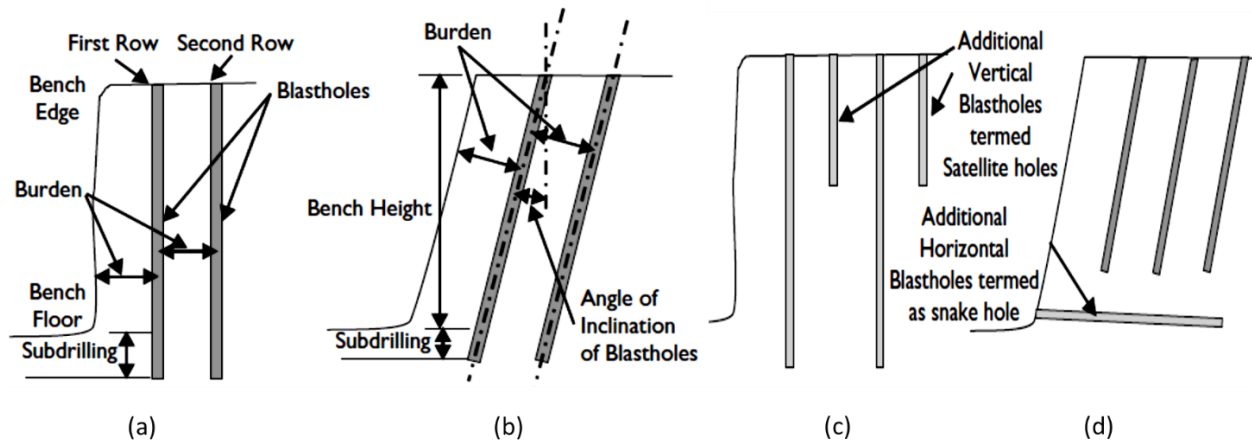


Figure 26: Vertical and inclined and additional blastholes in a bench

TYPES OF BLASTING IN SURFACE MINES

Blasting activities in large mines are carried out for breaking the rock mass, but the associated objectives are different for each one of them. This gives rise to different types of bench blasting techniques.

Conventional bench blasting: Conventional bench blasting is also called production blasting. In this type of blasting the purpose of the blast is to break a large portion of rock mass to yield such fragment sizes that can be easily loaded by using rope shovels or hydraulic shovels into the dumpers. These dumpers move the overburden to a waste site and the ore to a crusher site or store yard.

In the year 1972 Konya and Walters proposed an equation for calculation of burden as under,

$$B = 3.15 * D * (p_e/p_r)^{0.333}$$

where

B = Burden in ft

D = Diameter of explosive in inches

p_e = Specific gravity of explosive in g/cc

p_r = Specific gravity of rock in g/cc

On the basis of burden distance arrived at by using the above equation, the spacing distance can also be calculated as per following equations suggested by Konya and Walters.

For instantaneous blast of single row blastholes

$$S = (H + 2B)/3 \text{ when } H < 4B$$

$$S = 2B \text{ when } H > 4B$$

For sequenced blast of single row blastholes

$$S = (H + 7B)/8 \text{ when } H < 4B$$

$$S = 1.4B \text{ when } H > 4B$$

In all the above equations

B = Burden calculated as above in ft,

S = Spacing in ft

H = Bench height in ft

Stemming distance to be kept as under,

For massive rock $T = B$

For Stratified rock $T = 0.7 * B$

In the year 1983, Konya and Walters refined their approach and came up with the following equations.

$$B = ((2\rho_e/\rho_r) + 1.5) * D$$

$$\text{Stemming distance } T = 0.7B$$

$$\text{Subdrilling distance } J = 0.3B$$

Secondary blasting: The loading and moving equipment used in mines has limitations. Pieces of rocks beyond a certain size can neither be lifted nor carried. Even the crusher cannot accept rock pieces beyond a certain size for further crushing. Economic reasons always force the blast designer to design the blast in such a way that the largest fragments yielded through the blast are of a size acceptable to the lifting or moving equipment. Due to so many uncontrollable factors in blasting, in some cases large rock pieces are left even by a well-designed blast. Such large pieces can be moved only after their further blasting. Such blasting of large rock pieces is called secondary blasting.

Presplitting: In simple words presplitting means creating a fully cracked surface between the area to be blasted and the area to be kept intact. Such a surface is created by charging closely spaced, small diameter blastholes drilled in a line and blasting them instantaneously.

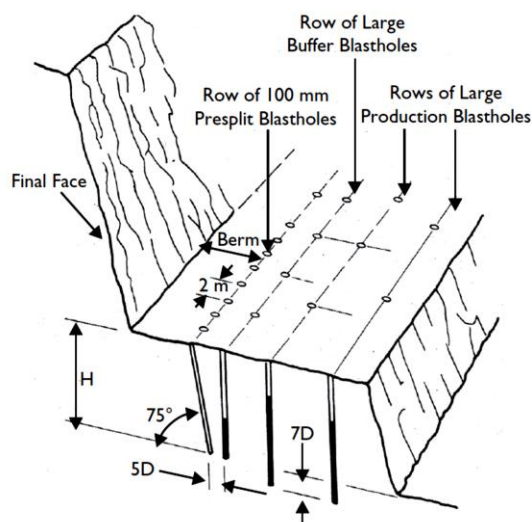


Figure 26.1: Presplitting behind the main bench

Diameters of blastholes for the presplitting operation is of the order of 100 to 125 mm. The most common method of drilling such blastholes is by rotary percussion drilling through hydraulic drifters because it gives the fastest penetration rates. Usually the presplit surface is at an angle of 15° with the vertical as shown in Figure 26.1.

The small diameter of the blastholes and the close spacing ensure that a well-trimmed smooth surface is formed without any damage on the either side of the blasthole plane. The presplit surface does not allow significant transmission of shock waves from the main blast into the rock mass meant to be kept intact. To meet this objective, often a row of buffer blastholes is also drilled in front of the presplit blasthole line. Blastholes in this row are charged to evolve limited energy.

Ideally a single fracture connects adjacent blastholes, and half of the hole remains at each presplit hole. If excessive crushing and radial cracking is observed at the plane it indicates excessive use of explosive. Closer spacing with an appropriate quantity of explosive gives a better surface. However, because of the high cost of drilling, the optimum spacing is the largest at which radial cracks will join and form a continuous undamaged surface. Smooth blasting is also a type of presplitting where the blastholes are of much larger diameter, often drilled with the same drill that is used for drilling production blastholes in the main bench.

Cast blasting: In many places coal is found in horizontal layers that underlie a large bed of overburden. In such cases the most economical method to reach the coal is by blasting the top overburden layer in the form of a long strip and moving it to one side by using a giant walking dragline as shown in Figure 27. In such mines the technique of cast blasting can be adopted with advantage.

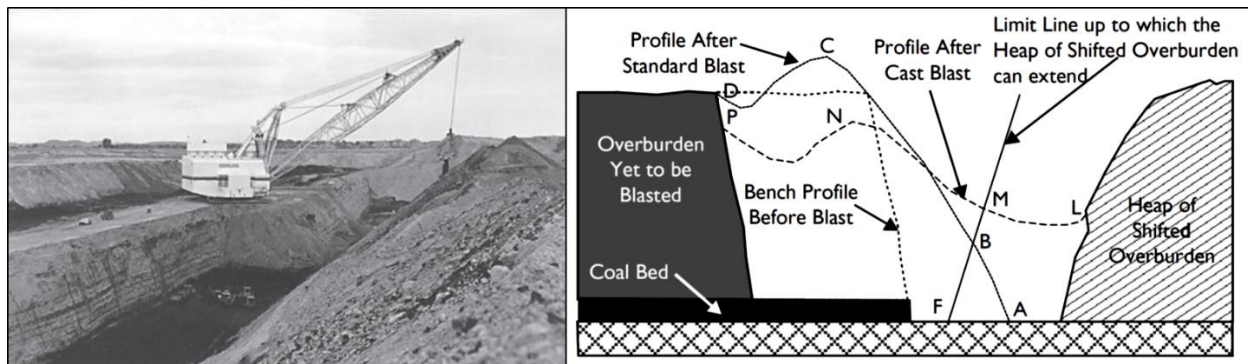


Figure 27: Photograph of a working dragline and sketch of profile of a castblasting

The advantage of cast blasting can be appreciated by means of the sketch in Figure 27. If the bench is blasted in the normal way, the profile of the top of the muck heap reaches points ABCD. Here the blasted overburden in the area ABF has already moved to the area of heap of shifted overburden without use of the dragline. Similarly, if the bench is blasted by using the technique of cast blasting the top of the muck heap reaches points LMNP. In this case all the blasted overburden in the area LMF has already moved to the area of the heap of shifted overburden without use of the dragline. Since area LMF is much larger than area ABF cast blasting proves greatly advantageous by reducing the workload of the dragline.

For a successful cast blast the following parameters of the bench, pit, wall etc. should be practiced. The parameters are shown in Figure 28. Bench height should be more than 12 m. Pit width should be 1.25 times the height of the bench face.

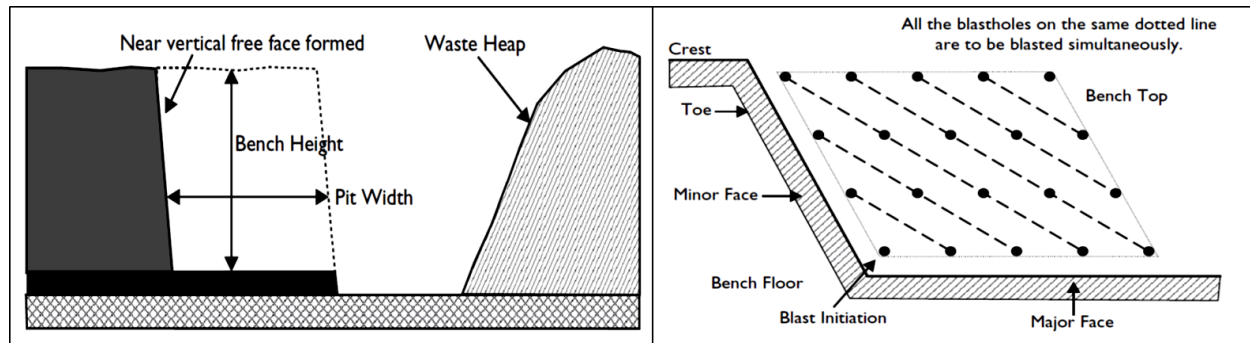


Figure 28: Bench parameter and firing sequence for dragline bench

Presplitting is essential because it enables the burden for the subsequent blast round to be the same for all the blastholes in the first row. Presplitting blastholes should be 251 mm dia., with spacing of 3 m in soft or tough rocks, and 5 to 6 m in hard and brittle rocks. The type of formation matters in choosing the quantity of explosive. Since soft rocks fracture quickly and the gases formed in the explosion escape early, the throw in the soft rock masses is low. To compensate for this the quantity of explosive required in blasting soft rock is rather high. Hard rock masses require less quantity of explosive.

Burden also greatly depends upon geological factors of the rock mass. It is usually determined from a nomograph on the basis of total charge weight in a blasthole and rock blastability.

The explosive selected must be able to give sufficient ejection velocity so the broken fragments are thrown in the appropriate direction to the appropriate distance.

The ejection velocity can be calculated from the following empirical equation:

$$V_e = 1.14 \times \left(\frac{B}{0.07853} \times d_2 \times \rho_e \times PAP \right) - 1.17$$

where

V_e = Ejection velocity in m/s

B = Burden in m

d = Charge diameter in cm

ρ_e = Density of explosive g/cm³

PAP = Absolute weight strength in cal/g

Minimum ejection velocity is about 15 m/s.

Spacing should be about 1.3 to 1.6 times burden. The blasthole layout pattern should be rectangular staggered. Stemming should be 20 to 25 times blasthole diameter. Subdrilling should be low i.e. 4 to 6 times blasthole diameter. This is essential so as to ensure that the coal layer is not broken, mixed and thrown with the overburden. Powder factor varies between 0.3 to 0.8 kg/m.

All the blastholes in a row should be fired simultaneously without any delay so long as only one face is involved. If two faces are involved then the firing sequence should be as shown in Figure 28. In order to ensure that material fragmented and thrown from the first row blast does not infringe with the material

fragmented and thrown by the second row blast, the delay timing between rows should be worked out on the basis of 20 to 25 ms/m of burden.